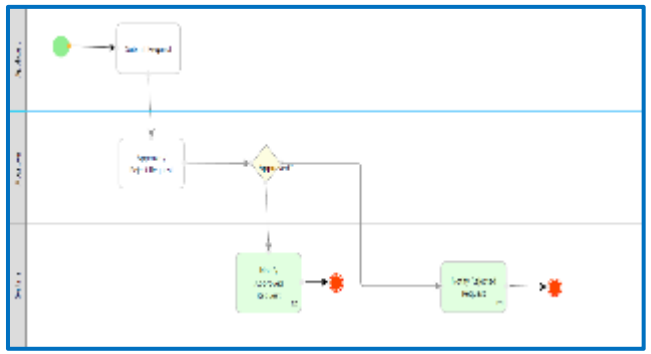
AAPP016-4-2
DevOps and Low Code Development (DLD)

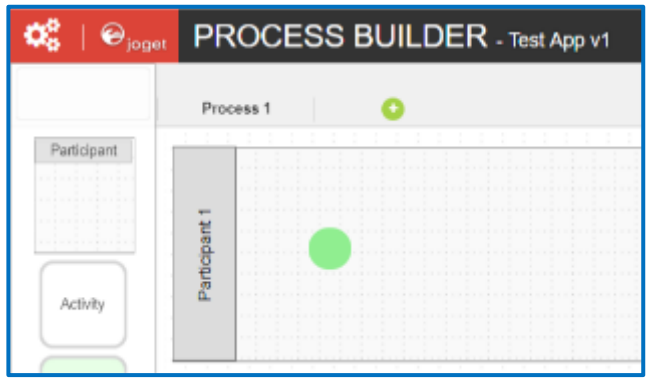
Lab Exercise - Translating Business Concepts into Applications

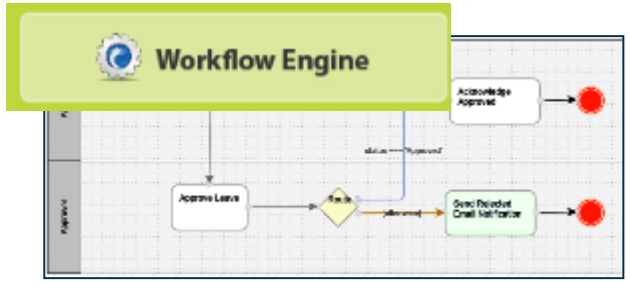
Designing your first Process with Process Builder using:

Joget®

Business Process Automation







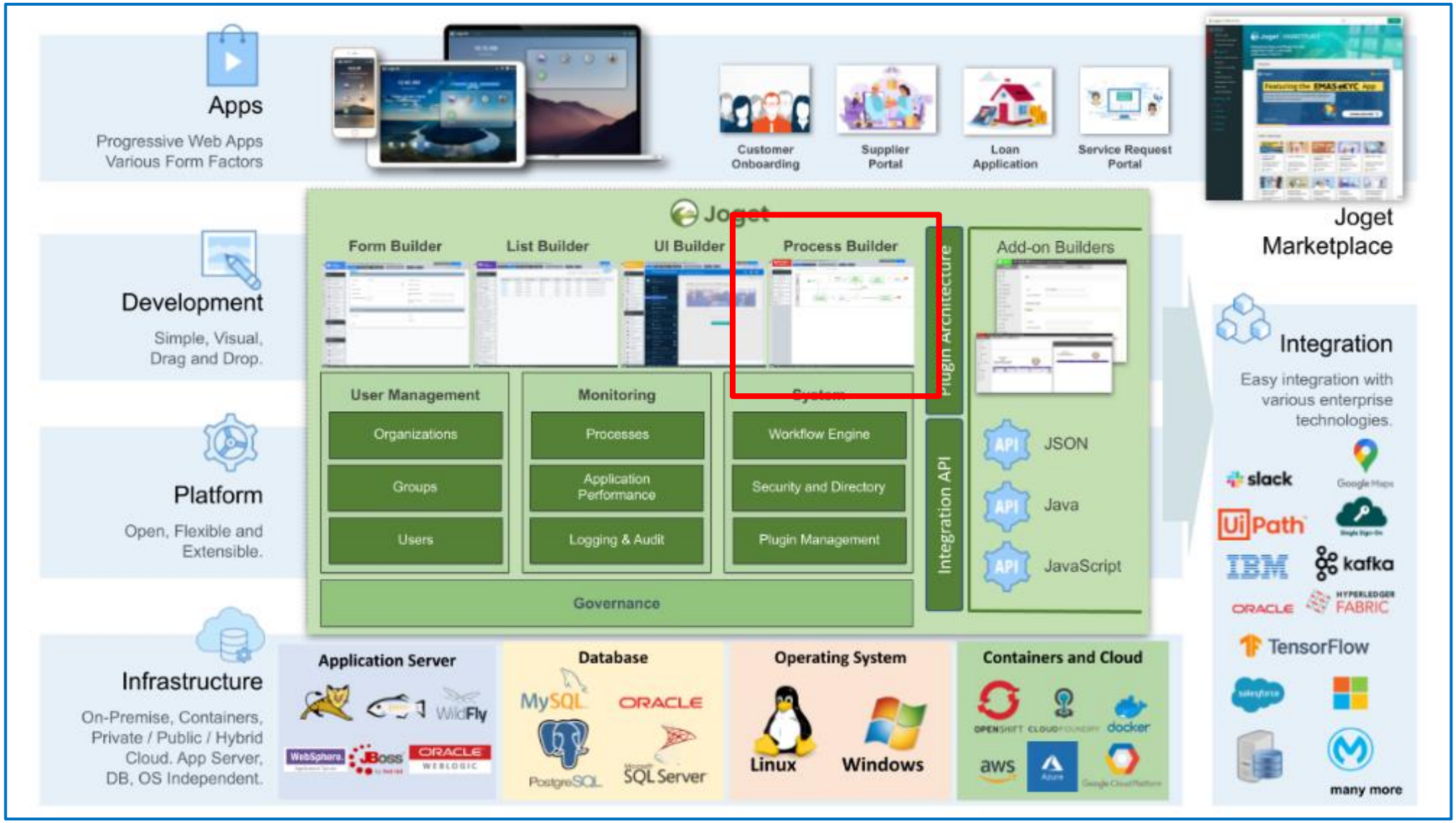
Business Process Design

↓

Process Design

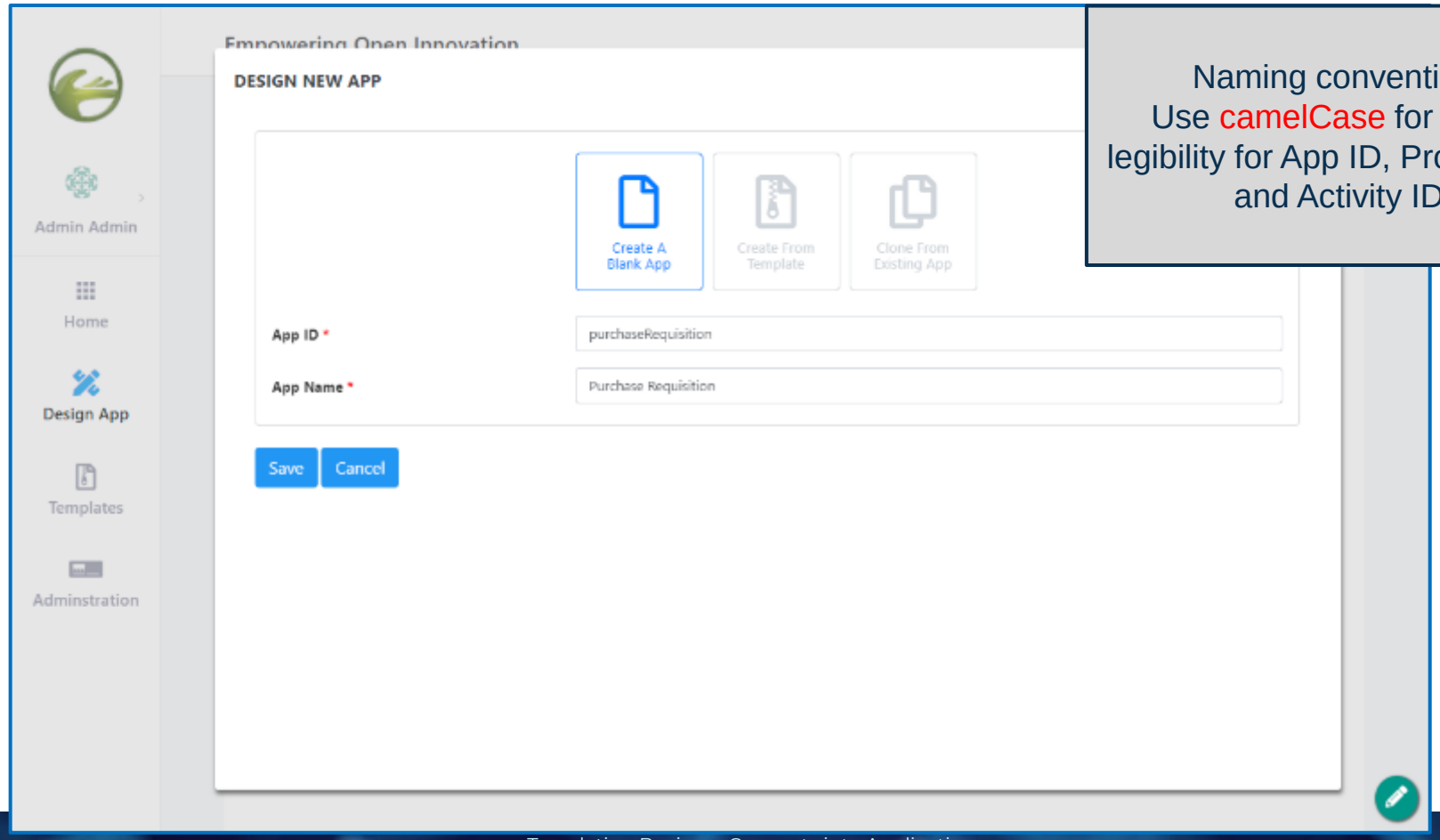
↓

Process


Apps
Progressive Web Apps
Various Form Factors
Customer Onboarding
Supplier Portal
Loan Application
Service Request Portal
Joget Marketplace
Development
Simple, Visual, Drag and Drop.
Joget
Form Builder
List Builder
UI Builder
Process Builder
Add-on Builders
Plugin Architecture
Integration API
User Management
Organizations
Groups
Users
Monitoring
Processes
Application Performance
Logging & Audit
System
Workflow Engine
Security and Directory
Plugin Management
Governance
JSON
Java
JavaScript
Platform
Open, Flexible and Extensible.
Integration
Easy integration with various enterprise technologies.
slack
Google Maps
UiPath
IBM
kafka
ORACLE
HYPERLEDGER FABRIC
TensorFlow
many more
Infrastructure
On-Premise, Containers, Private / Public / Hybrid Cloud. App Server, DB, OS Independent.
Application Server
WildFly
WebSphere
JBoss
ORACLE WEBLOGIC
Database
MySQL
ORACLE
PostgreSQL
SQL Server
Operating System
Linux
Windows
Containers and Cloud
OPENSHIFT
CLOUDFOUNDRY
docker
aws
Azure
Google Cloud Platform

Create Your First App

- Design Apps -> Design New App



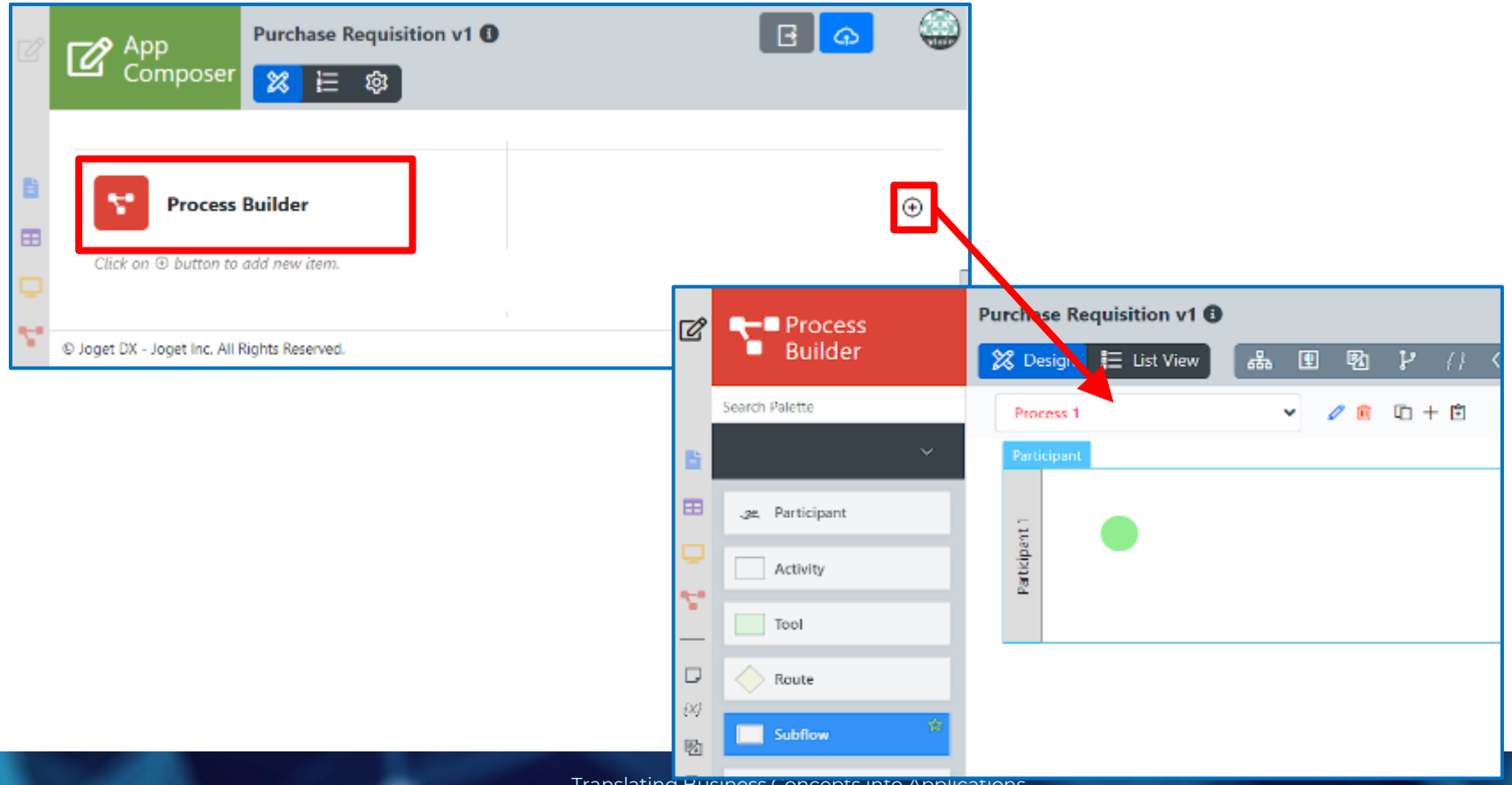
Naming convention:
Use camelCase for better legibility for App ID, Process ID, and Activity ID

Create Your First App

- Purchase Requisition Application
  - Purchase Request Process
    - Applicant **submits**
    - Approver **approves**
    - Applicant is **notified** of the request result

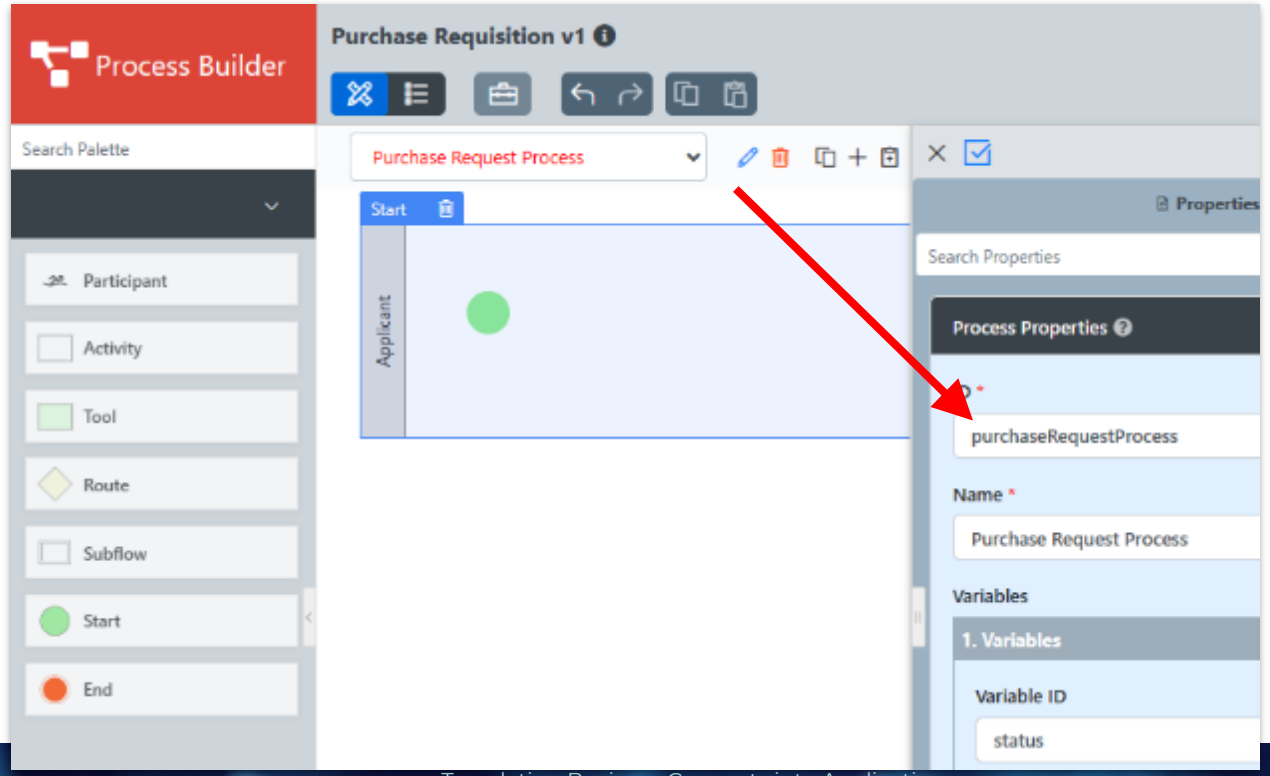
Design Your First Process

- Launch Process Builder – **App Composer > ⊕ (Add New) On Process Builder Section**



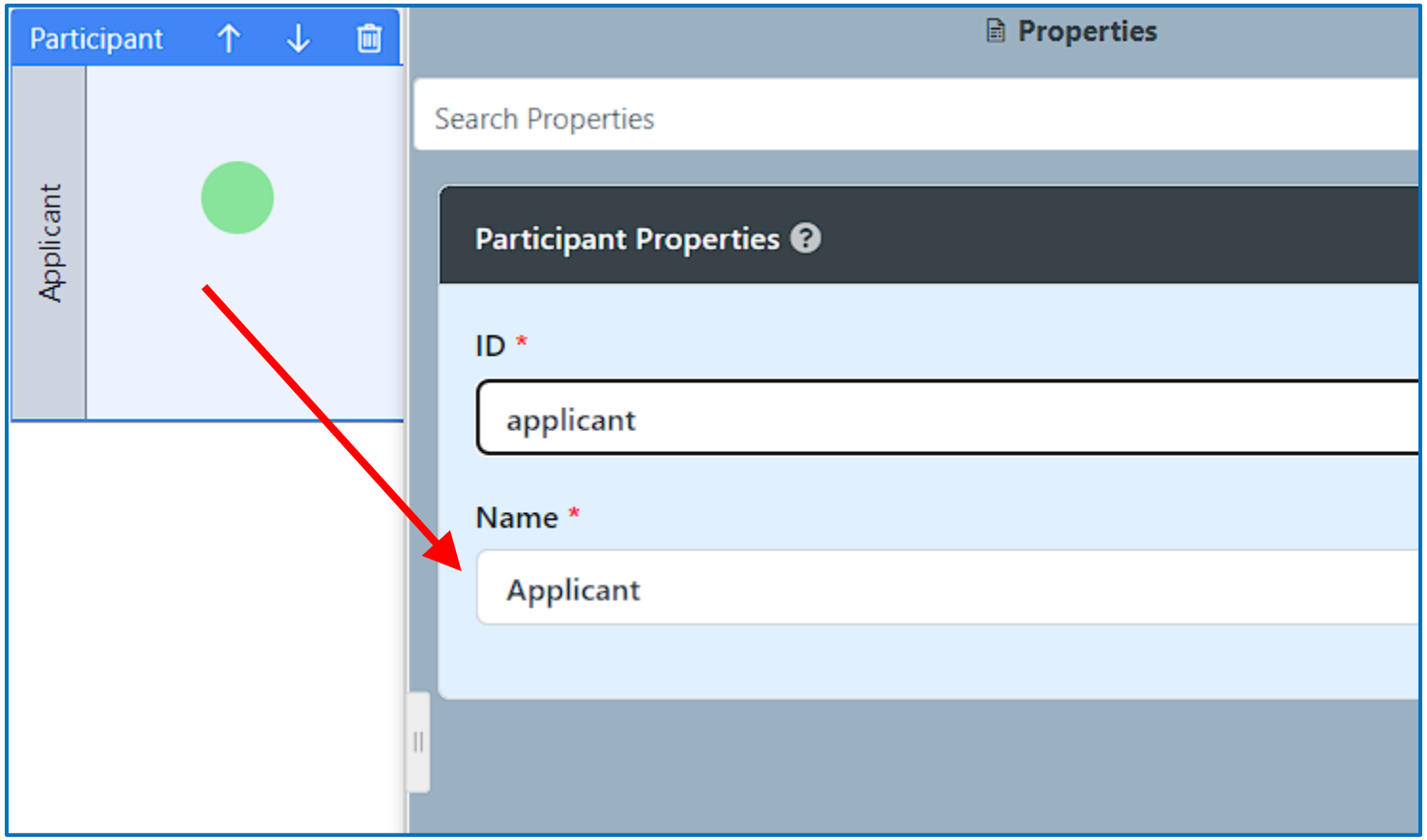
Defining Your Process

- Rename Process ID and Name
  - Use **camel case** for ID - **purchaseRequestProcess**
  - Do not use **snake case** as it conflicts with process instance ID naming convention



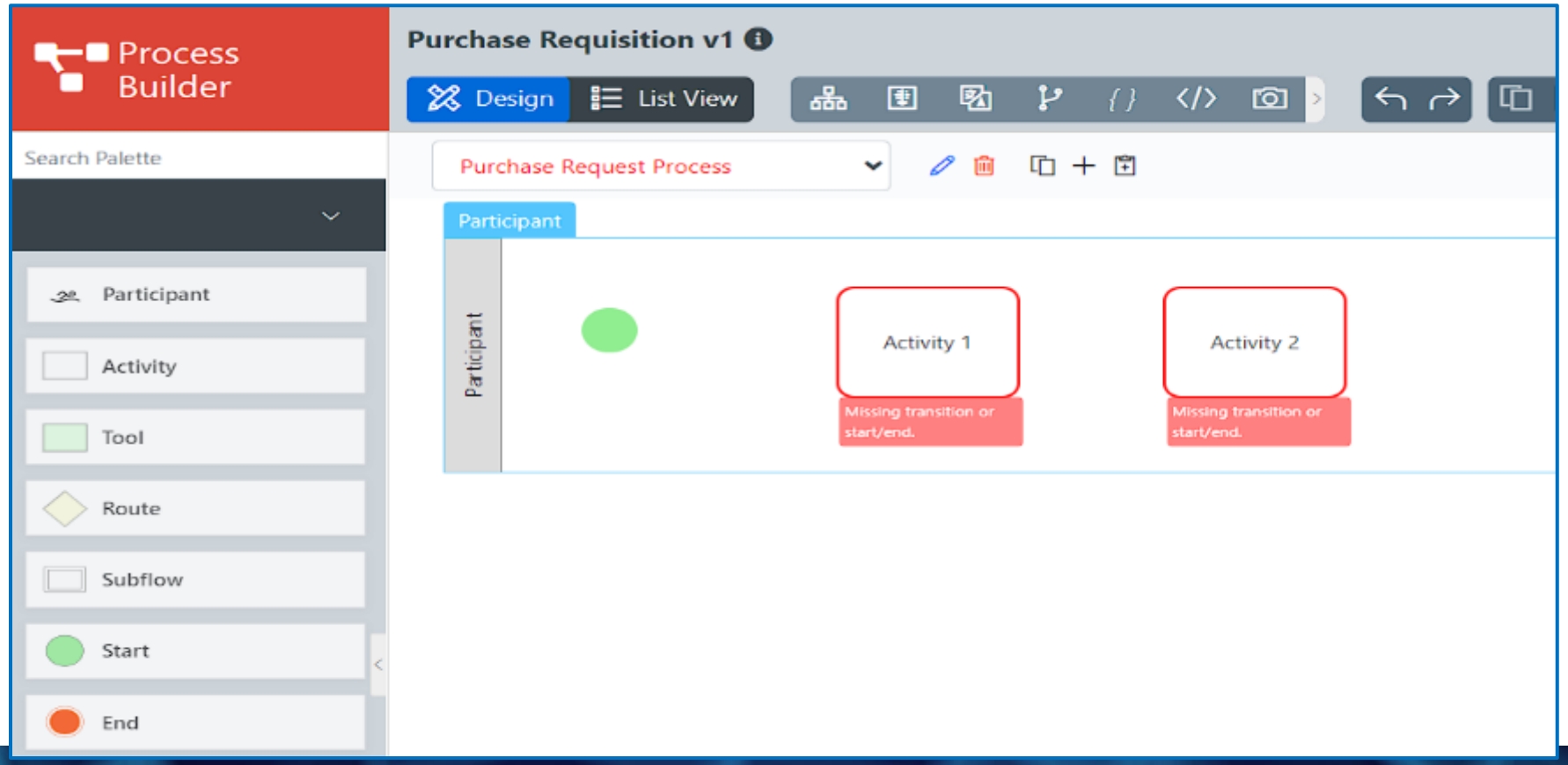
Defining Your Participant

- Rename the existing participant to **Applicant.**



Smart Guide

- The Process Builder will highlights logical errors as you design.

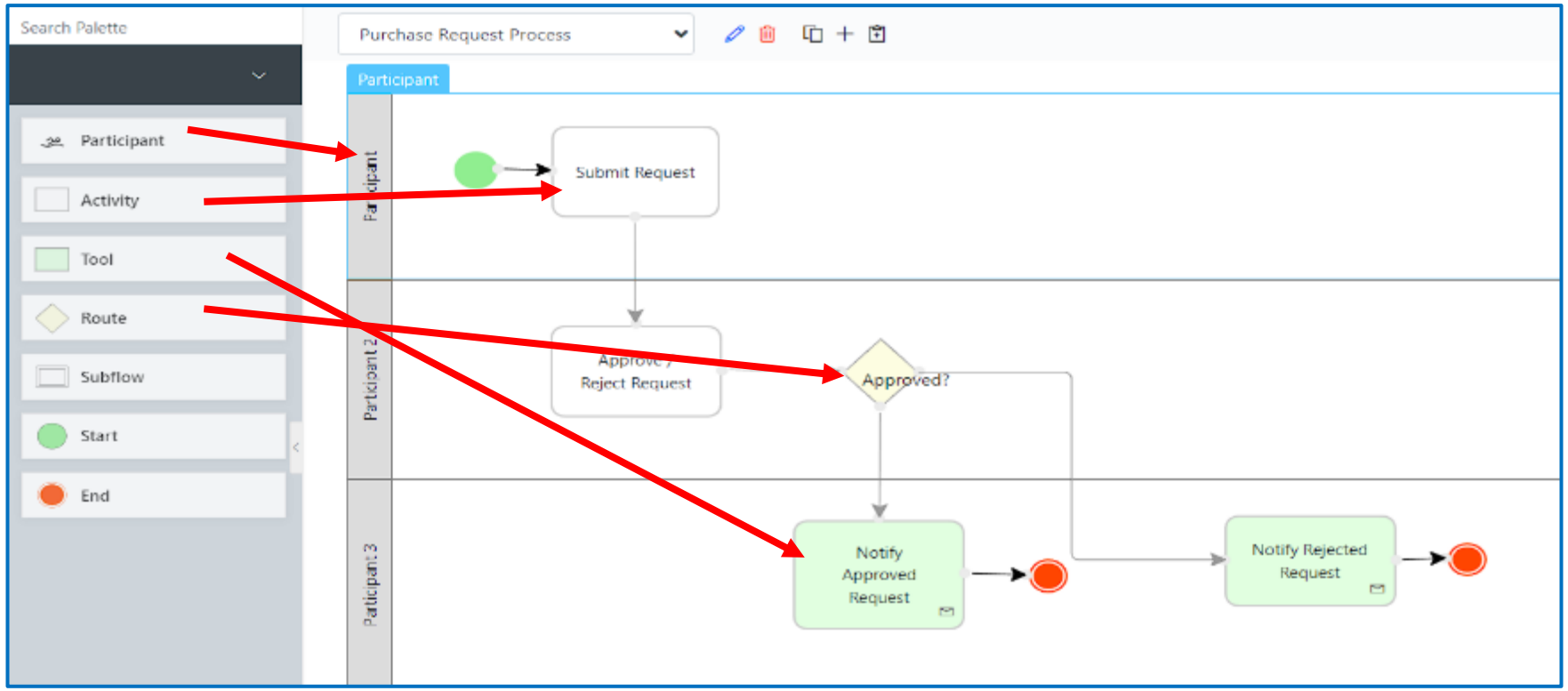


About Smart Guide

- Advises only on possible design errors.
- Will **NOT** advise on flawed business logic, design inefficiency.

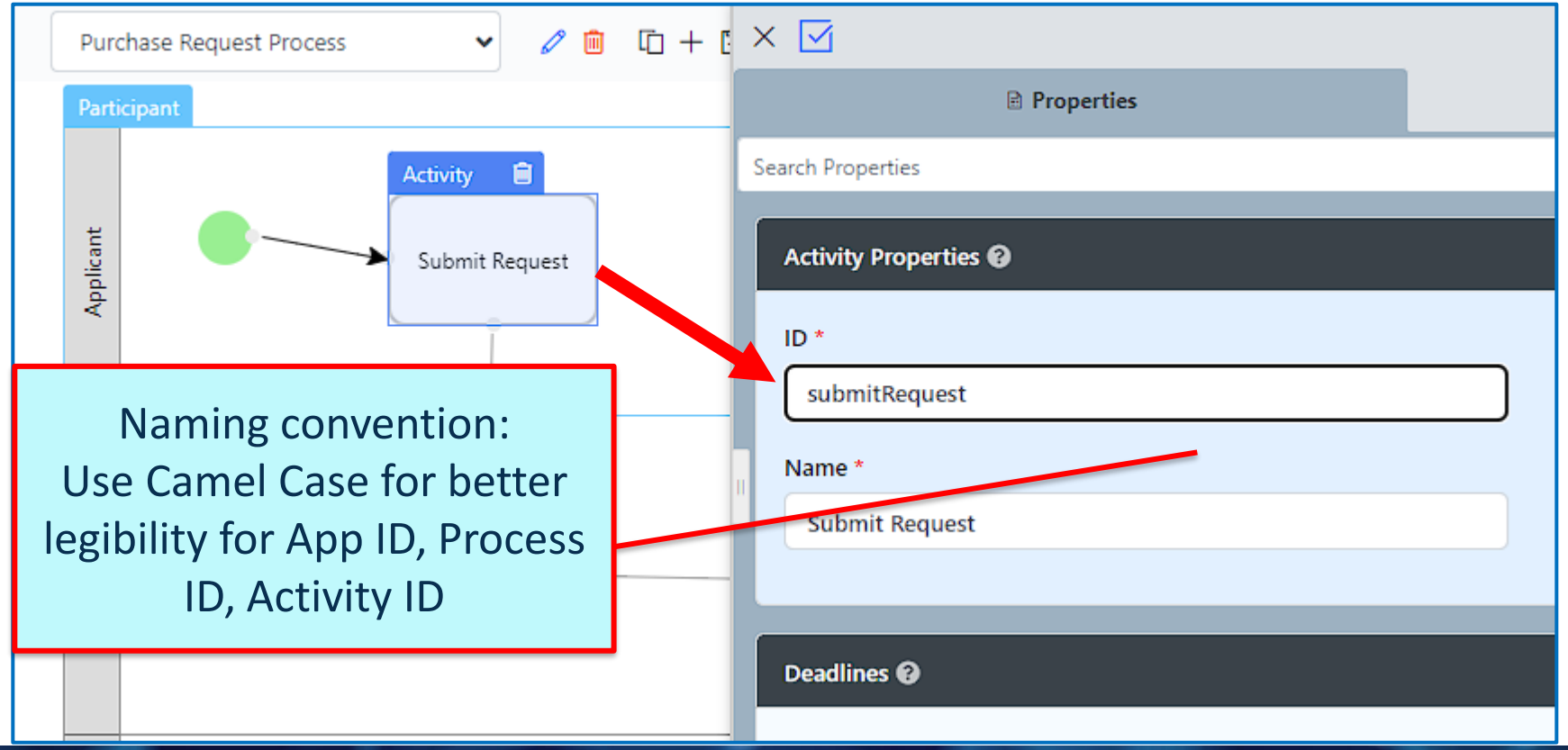
Insert Activities and Transitions

- Drag and drop, select the appropriate nodes to build the Process.



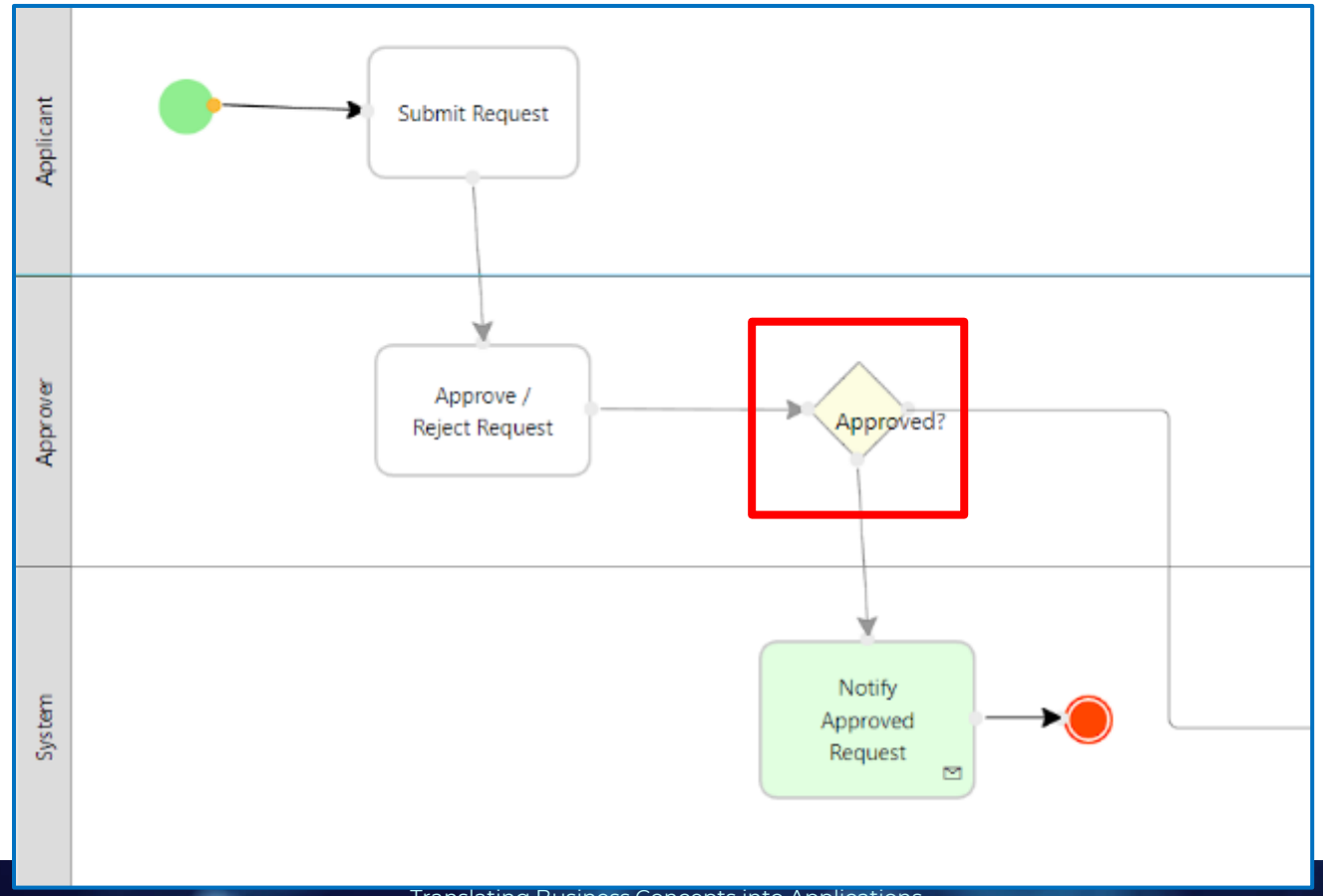
Rename Activity ID and Name

- Rename the activity node to make it meaningful.
- Use **camelCase** naming convention for ID (submitRequest).



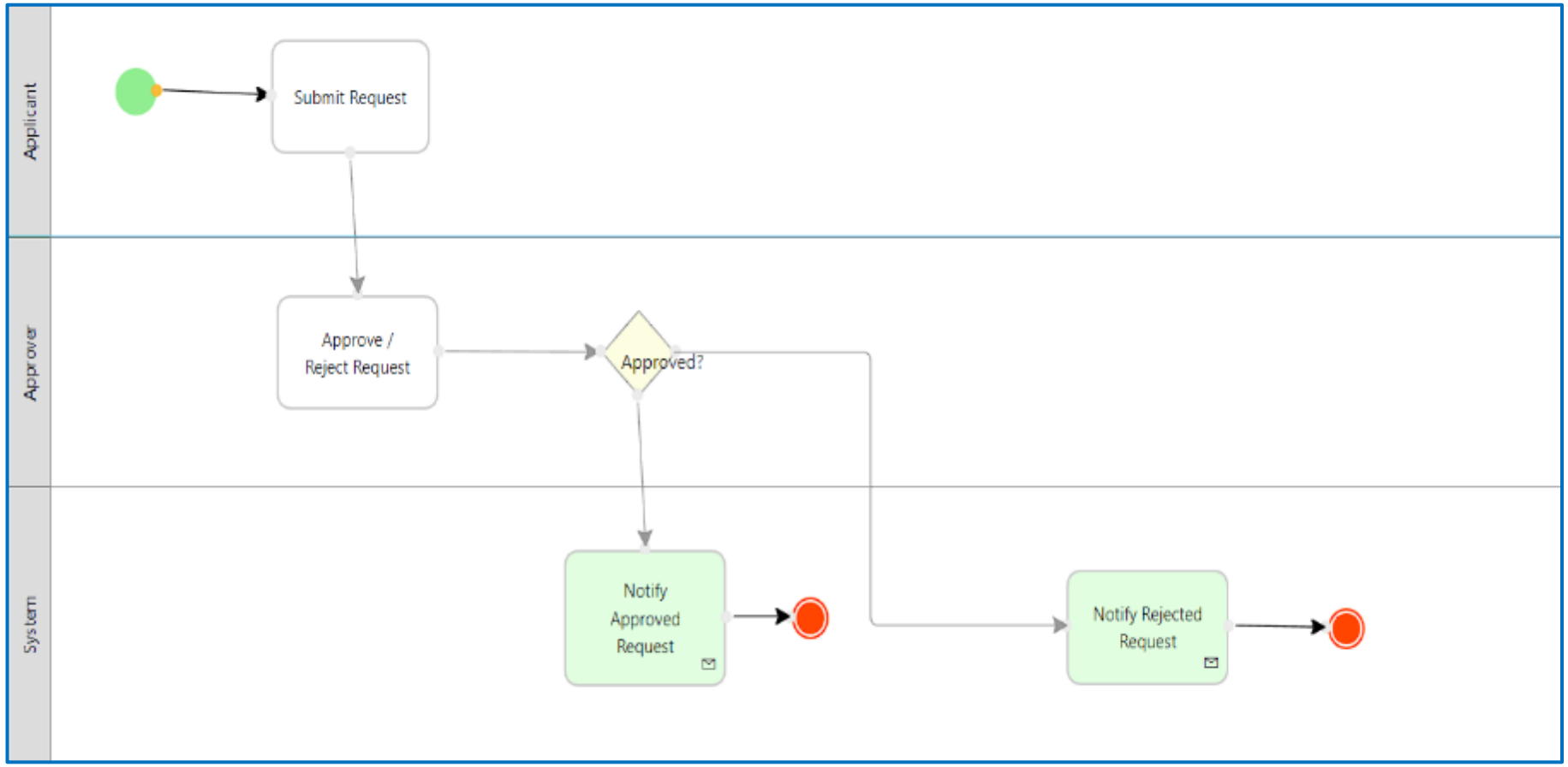
Implementing the Business Rule

- Route: Diamond-shaped node – Used to implement business rules.



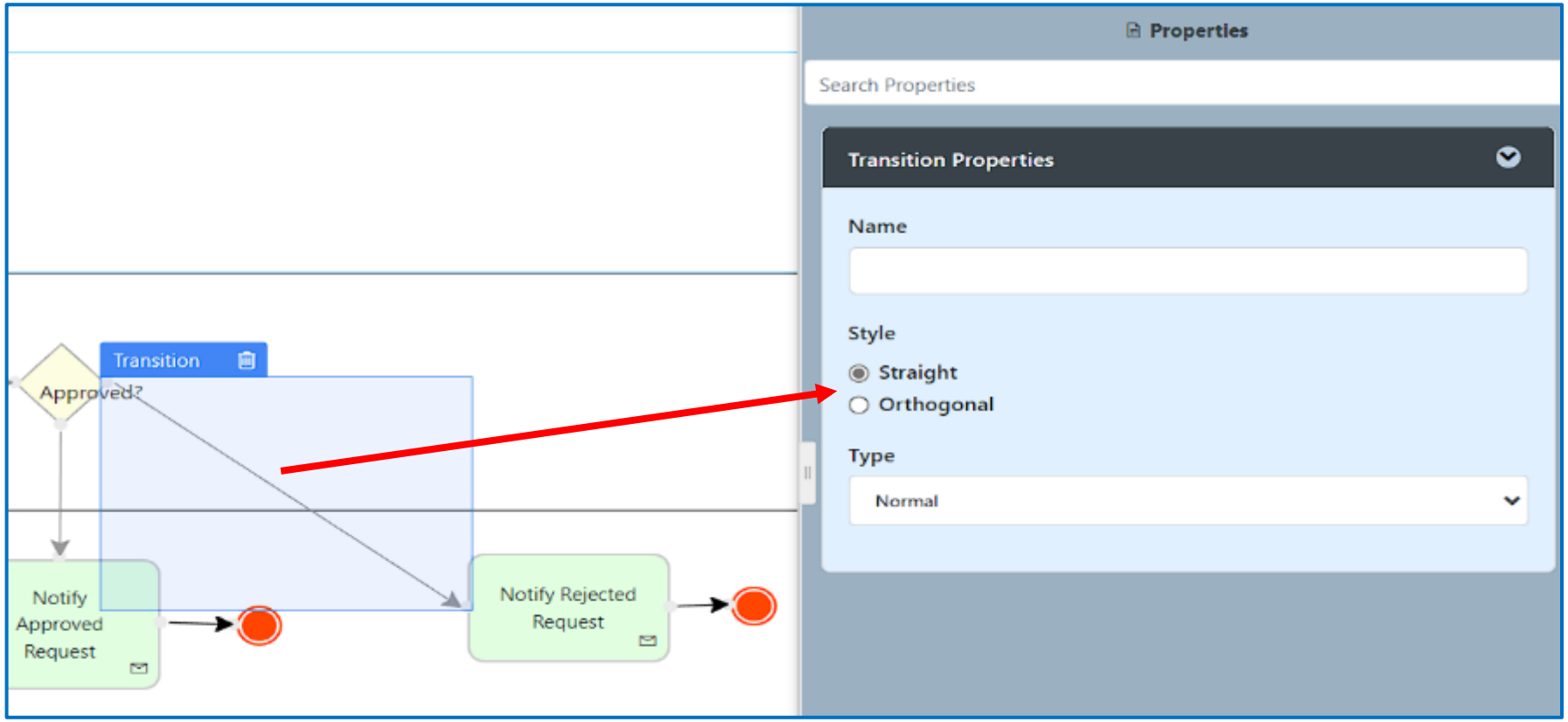
Completing the Design

- Insert the rest of the activities.



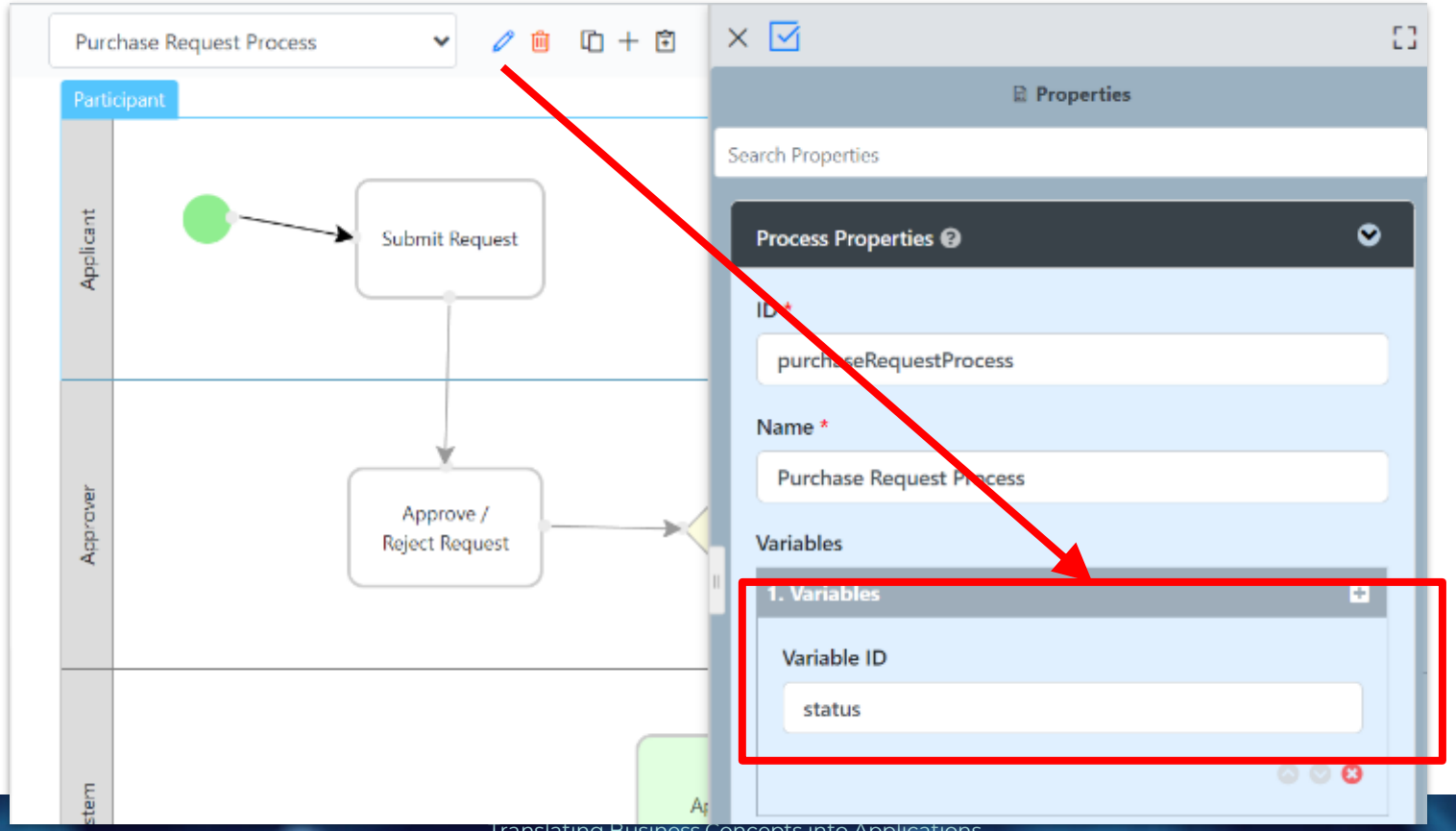
Managing the Transition Style

- Set style to Straight / Orthogonal to style your transition.



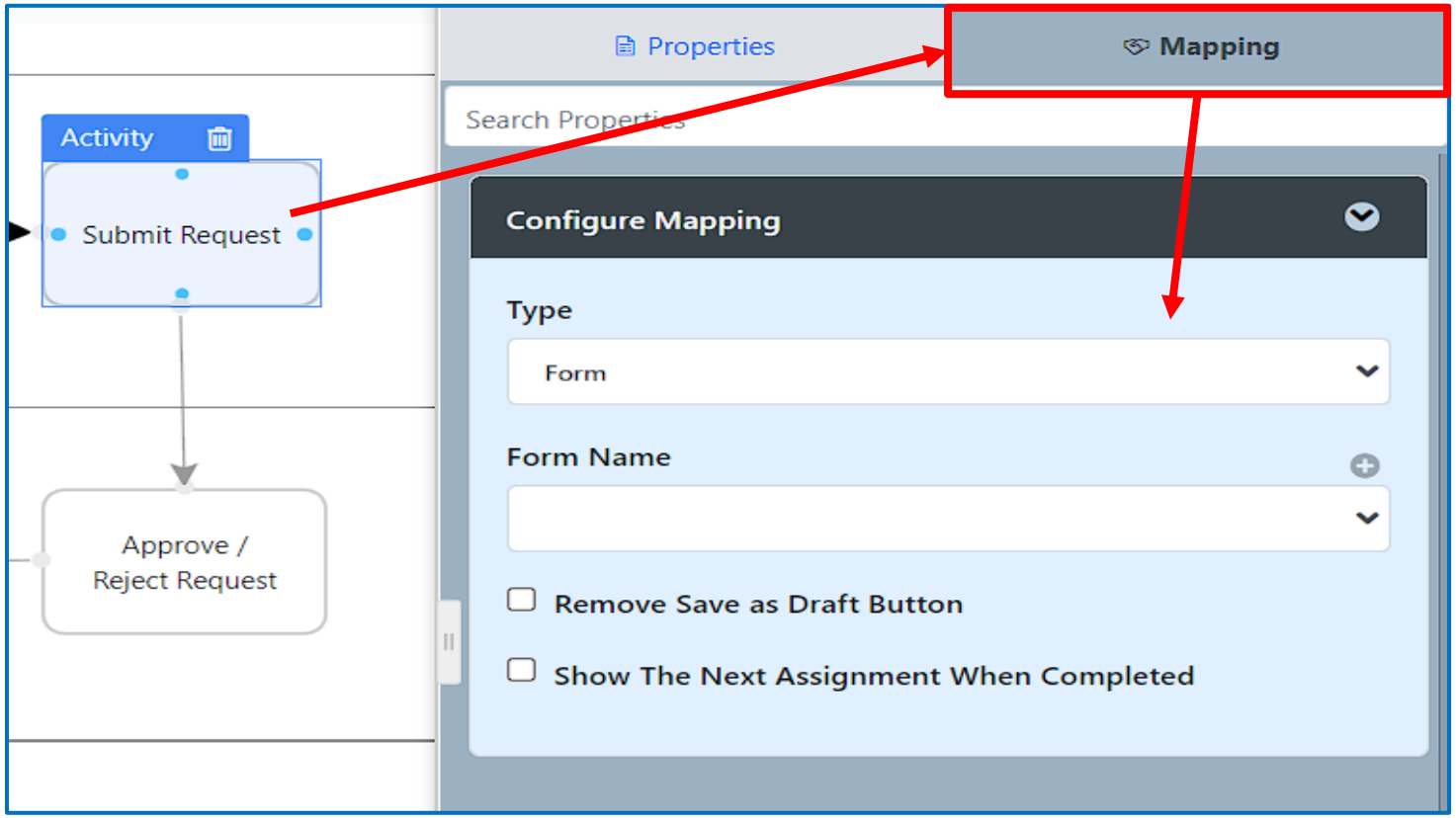
Declaring the Workflow Variable

- Variables used in the Process must be declared in the Process's properties.



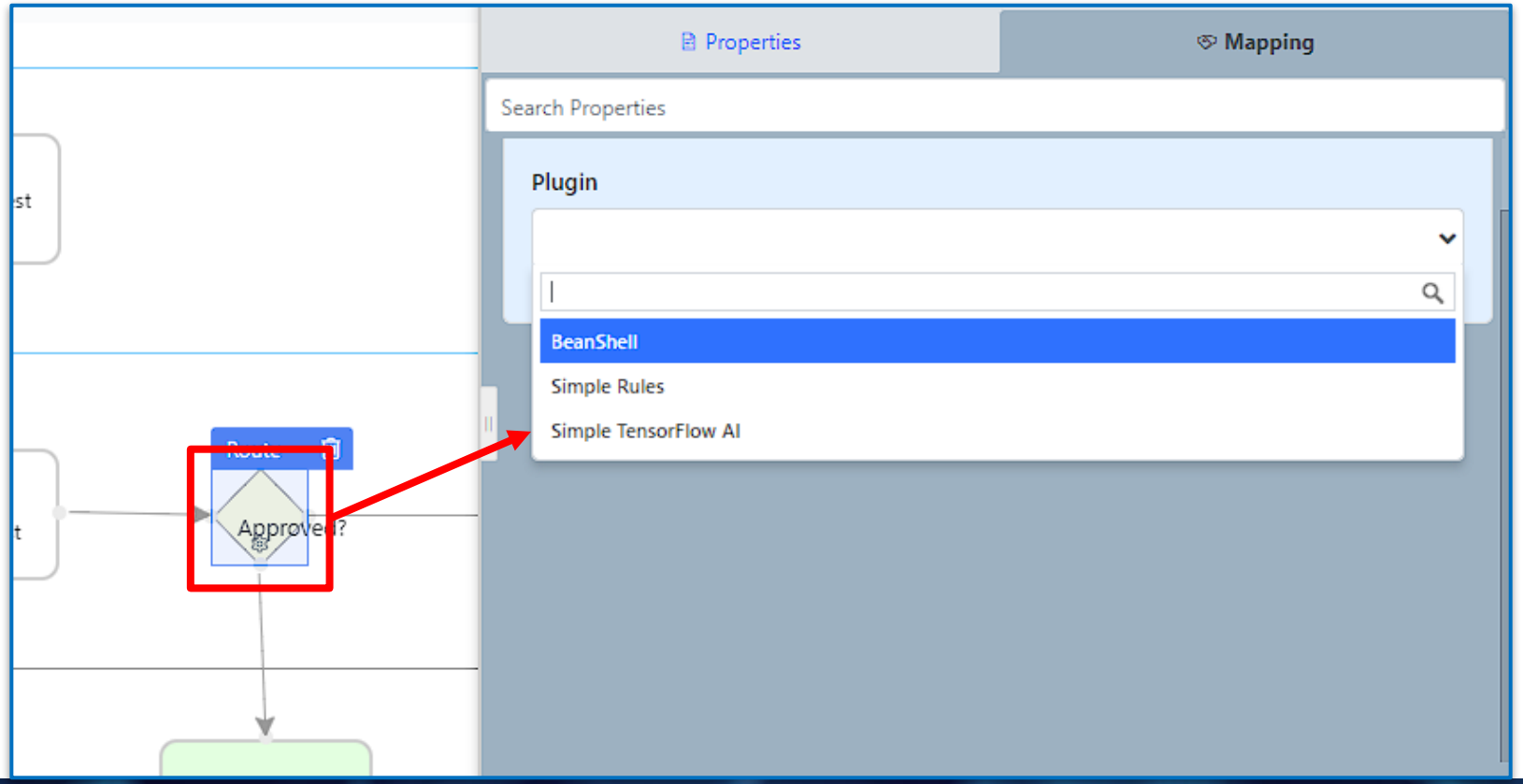
Implementing the Business Rules

- Process Mapper
  Process Builder > Any Node > Mapping Tab



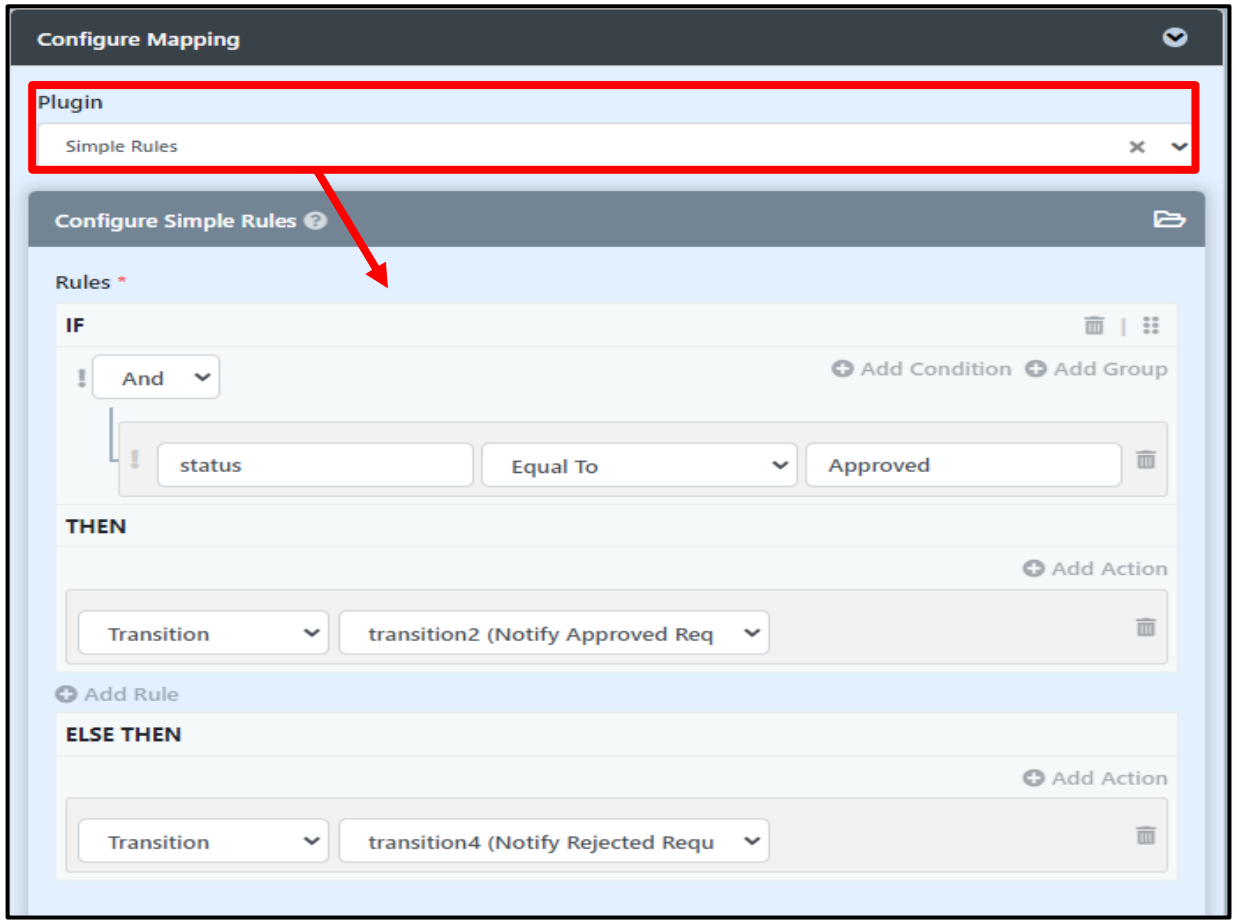
Implementing the Business Rules

- Route: Diamond-shaped node – Used to implement business rules.



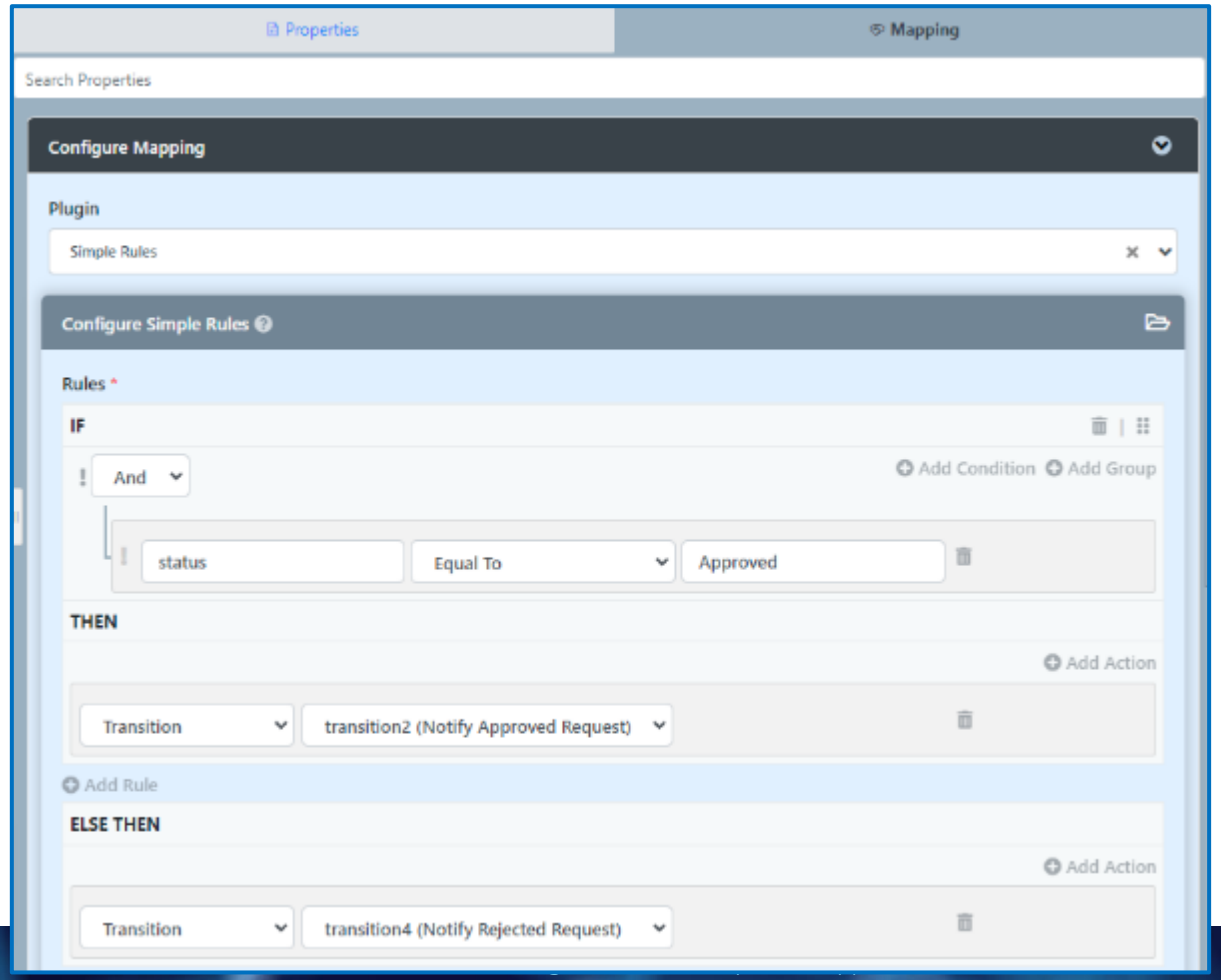
Implementing the Business Rules

- Choose **Simple Rules.**



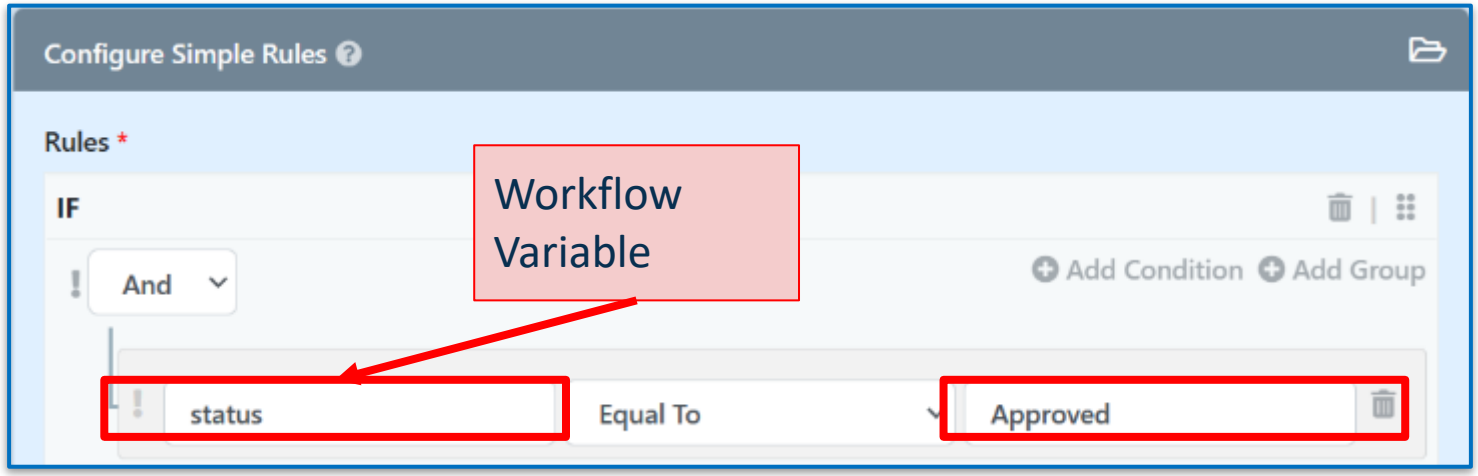
Setting the Rules for Transition Condition

- When the **status** variable is met with certain condition(s), route accordingly.



Simple Rules Decision Plugin

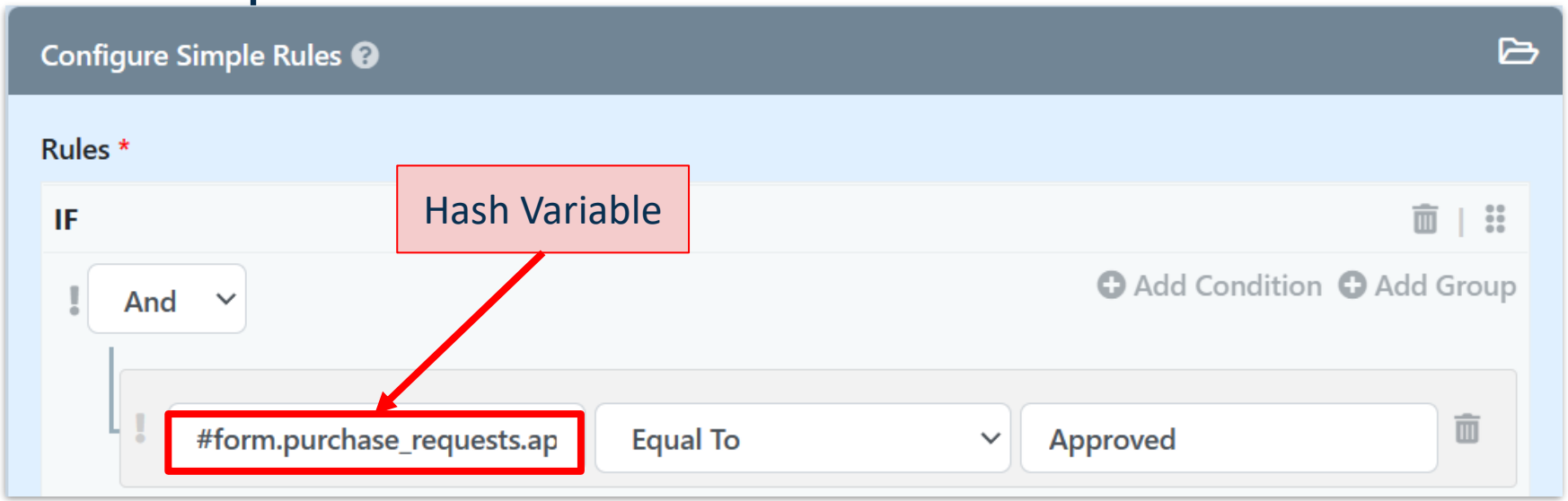
- **Hash variable** is supported in input fields as well for complex use cases.



Note: This means you don't even need workflow variables to control the route!
e.g.: using Form Hash variable to read form field value to route

Simple Rules Decision Plugin

- **Hash variable** is supported in input fields as well for complex use cases.

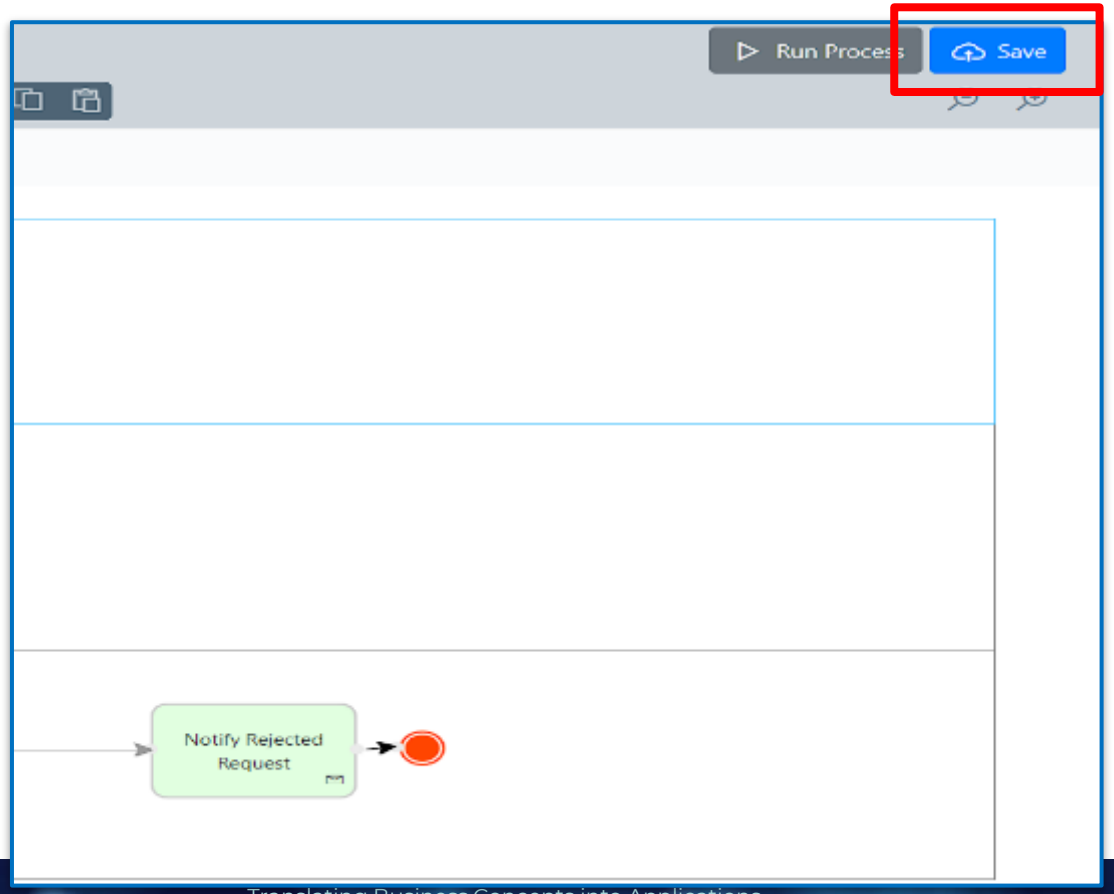


e.g.: using *Form Hash Variable* to read form field value to route

#form.purchase_requests.approval#

Deploying Your Process Design

- On the top right corner of the builder, click on the "Save" button to push your finished process to your Joget App.

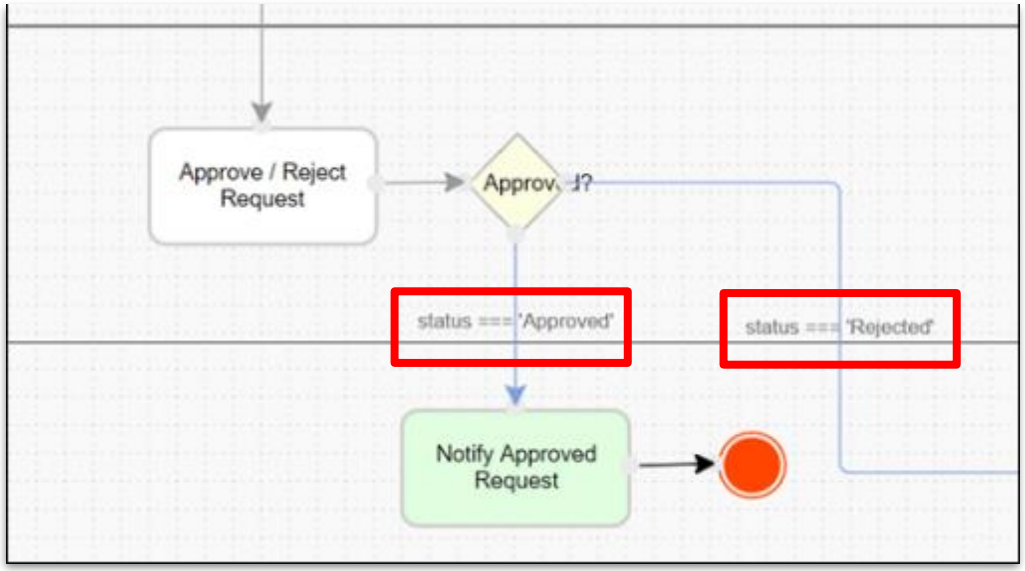


If and otherwise…

- Does this resemble the typical programming if-else condition block?
- Is this the best design?
- Why not if-else-if-else everything? (In other words, explicitly set all possible conditions)

Alternative Business Rule Implementation

- Prior to DX 8, business rules implementation must be done in the Process Builder.
- This approach is **no longer recommended** as it is difficult to change business logic without redeploying the whole process design of the entire app.



Advantages of using Process Mapper

- Decision Tool Plugin takes precedence over Process Builder conditional setup.

Good to Know

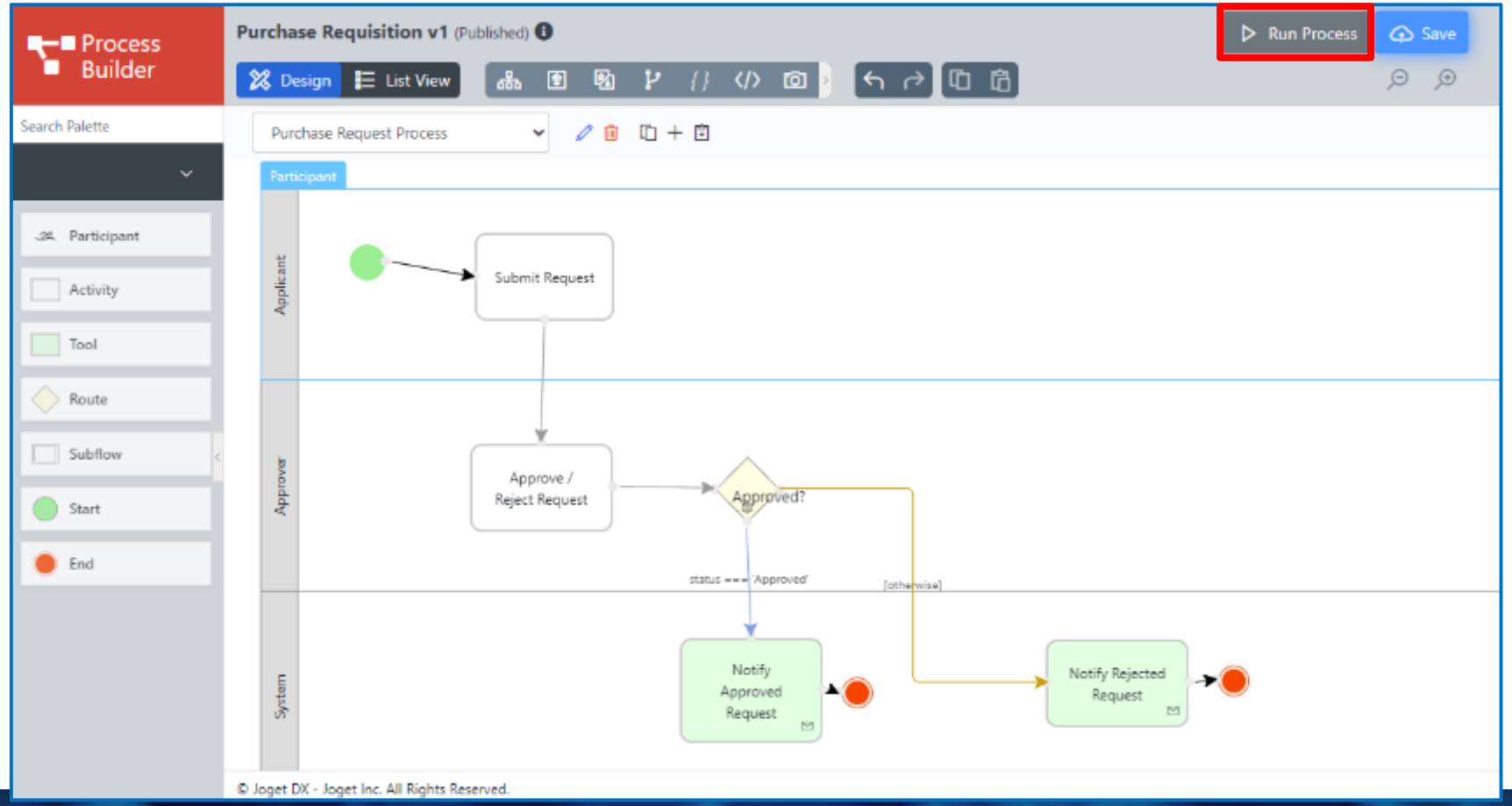
- Expressions in “Condition” are evaluated in JavaScript.
- Workflow variable are stored as data type **string**.

How to Test?

- There are 2 ways in general.
  - Make use of the "**Run Process**" button in the Process Builder
  - Make use of **UI**(for all users)

Verify the Process Using Run Process

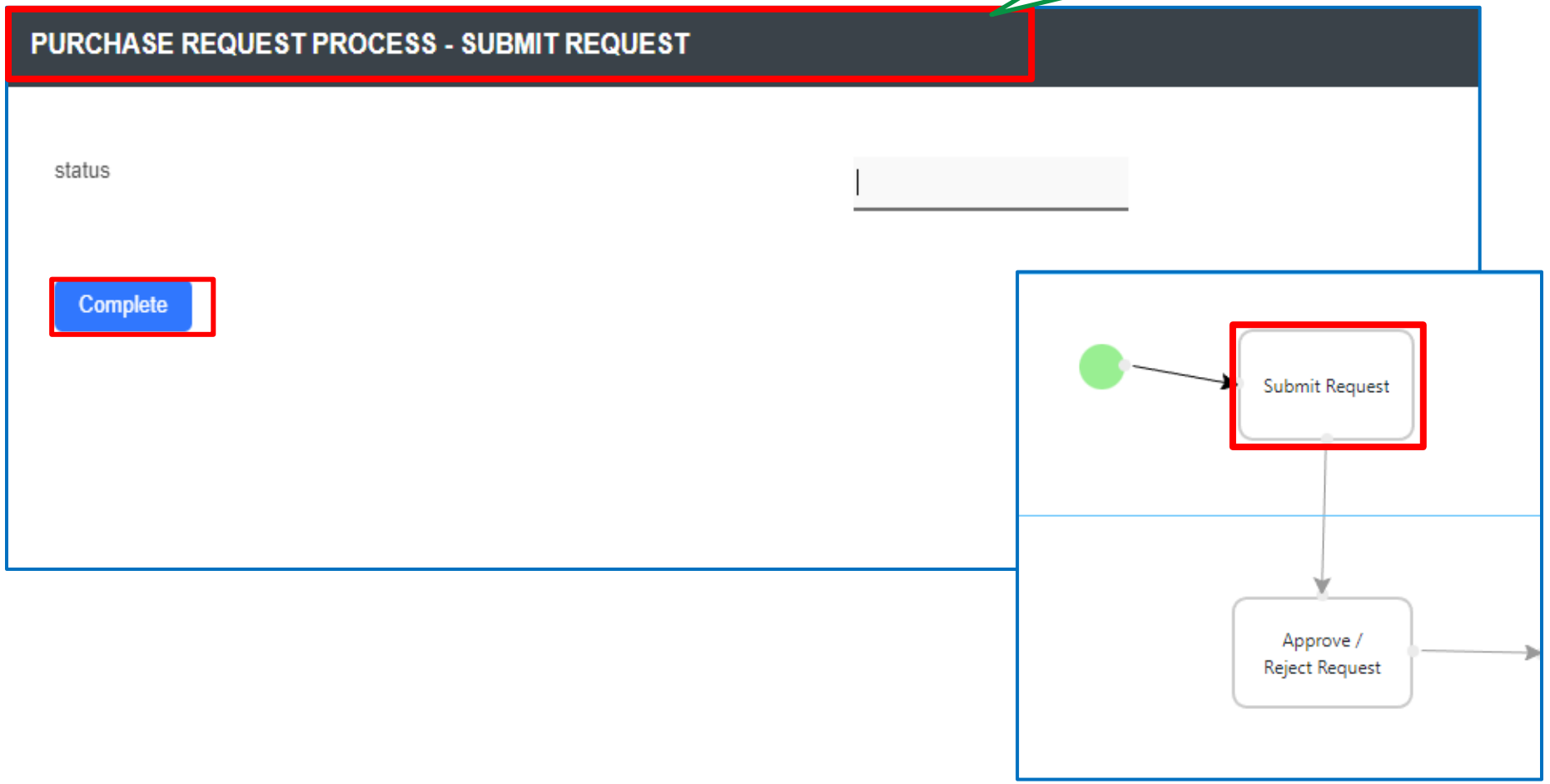
- Run through the Process to verify.



Verify the Process

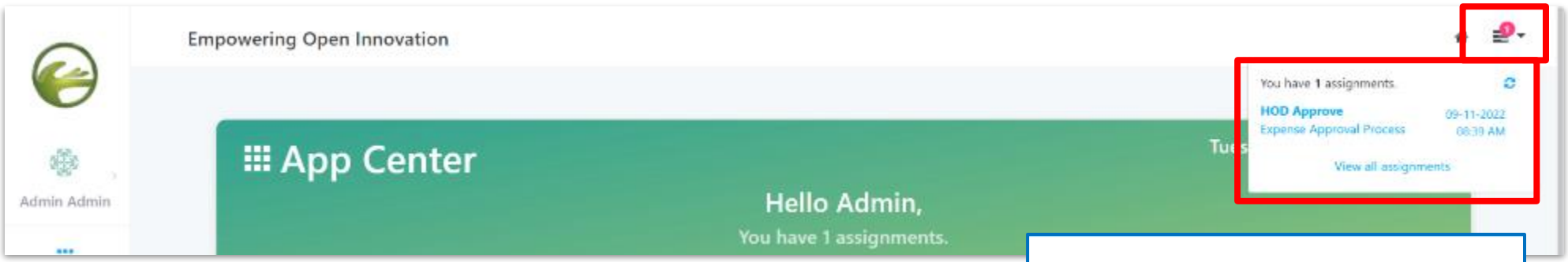
- Complete the assignment to proceed.

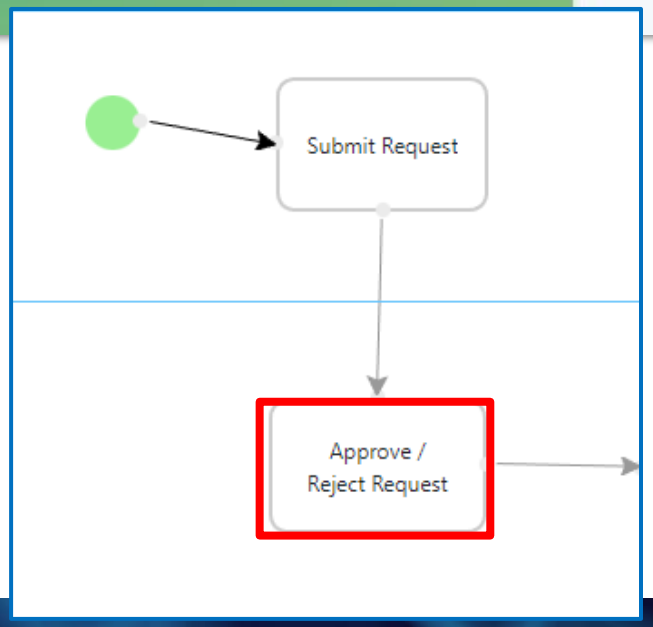
HINT: Activity name is configured in Process Builder



Accessing the Assignments

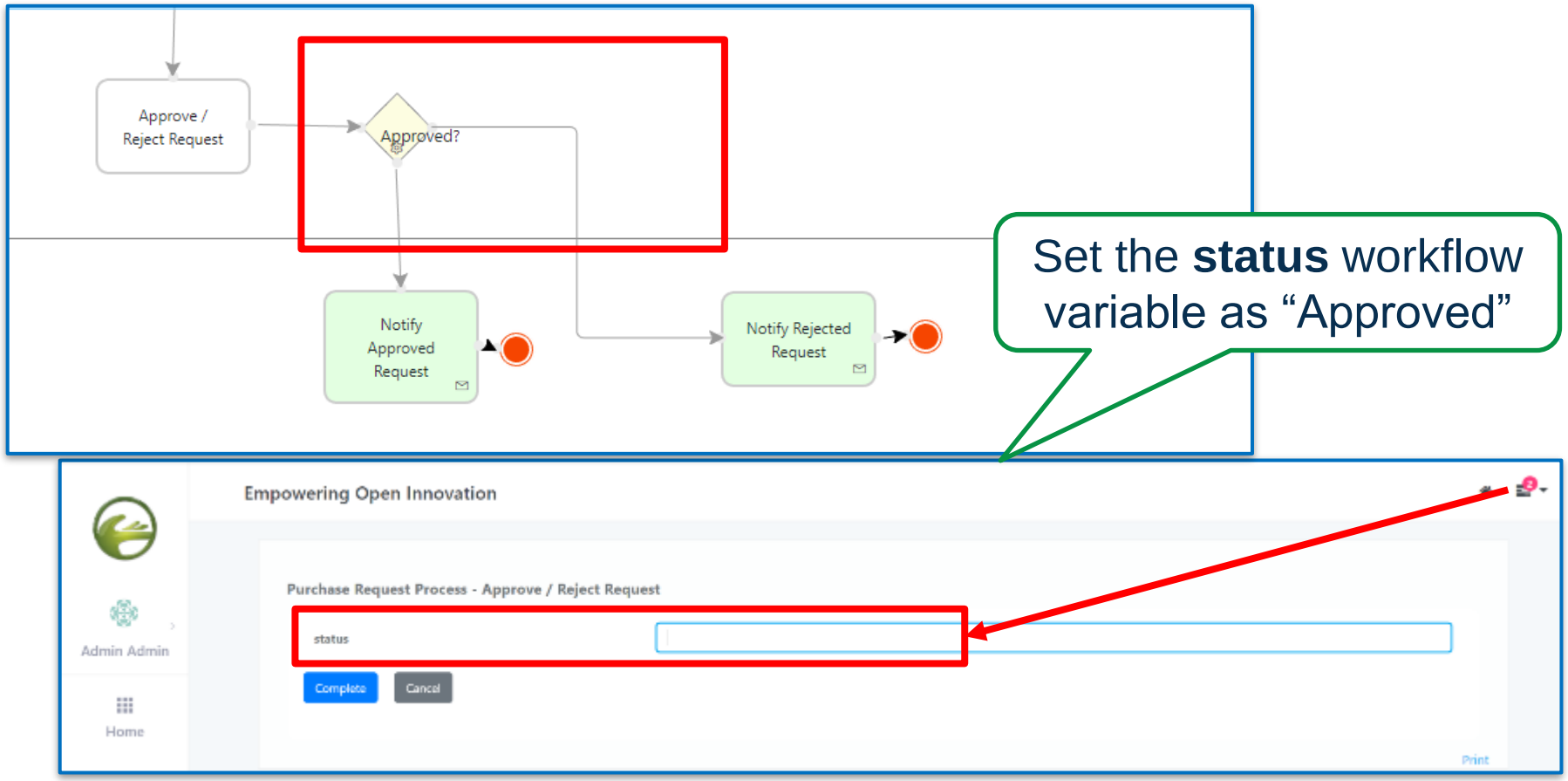
- App Center -> Universal Inbox





Determining the Flow Using Route

- Manipulate the Process using Workflow Variable and complete the process.



Exercise

- Run through the process from the start again to test out the “Rejected” case scenario.

Good to know

- At this point of time, the App is NOT published yet, therefore end user would not be able to run it.
- “Run Process” button at the App Designer’s Processes tab is a convenient way to verify your process flow.

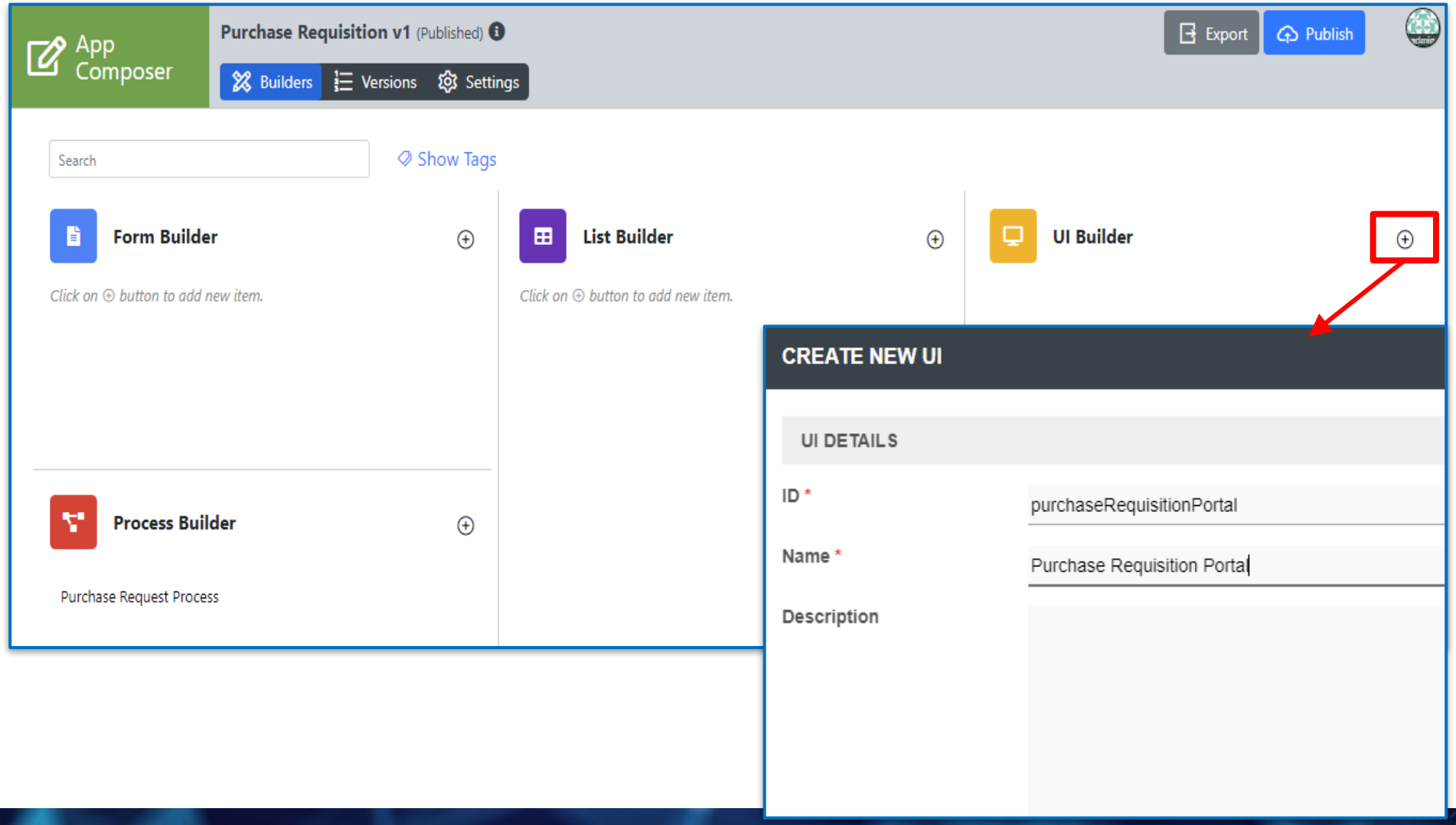
How to Test?

- There are 2 ways in general.
  - Make use of the "**Run Process**" button in App Design > Processes (only for admin)
  - Make use of **UI**(for all users)

Verify the Process Using UI

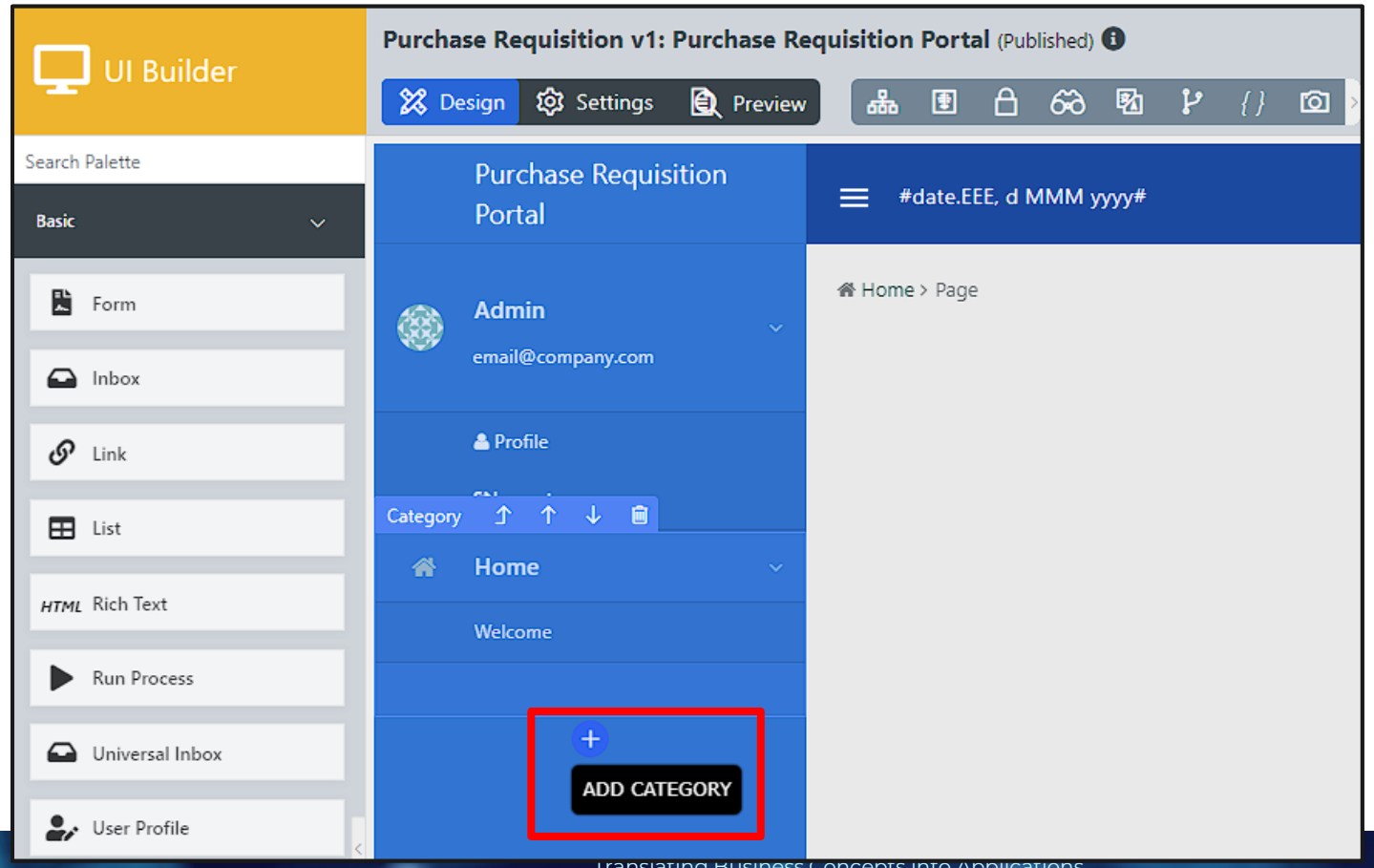
- Create a new UI.
- Add **Run Process** and **Inbox** element.
- **Publish** the App.
- Try running the whole process again by accessing the UI from the app center.

Create New UI



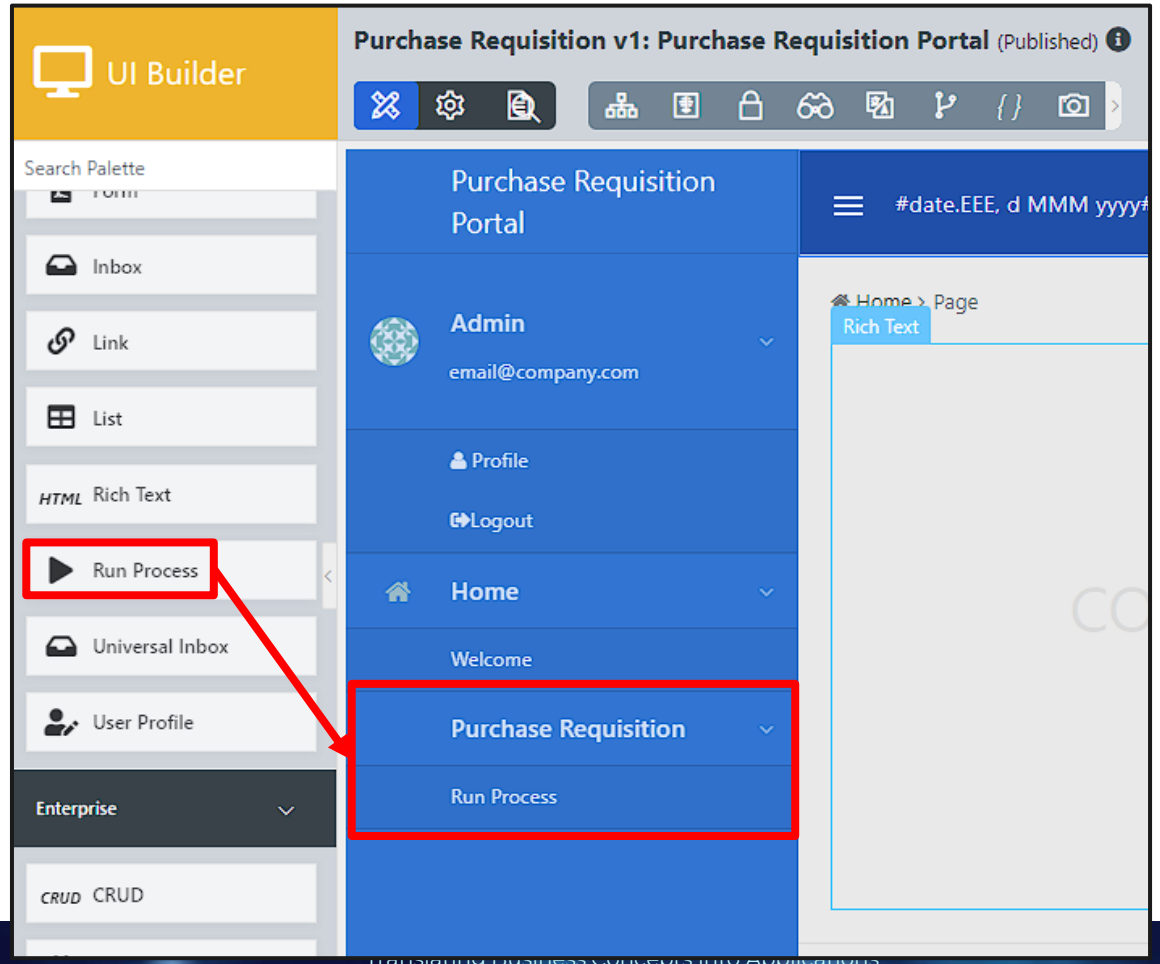
UI Category

- Items are organized by Category.
- Add 1 new category – Purchase Requisition.



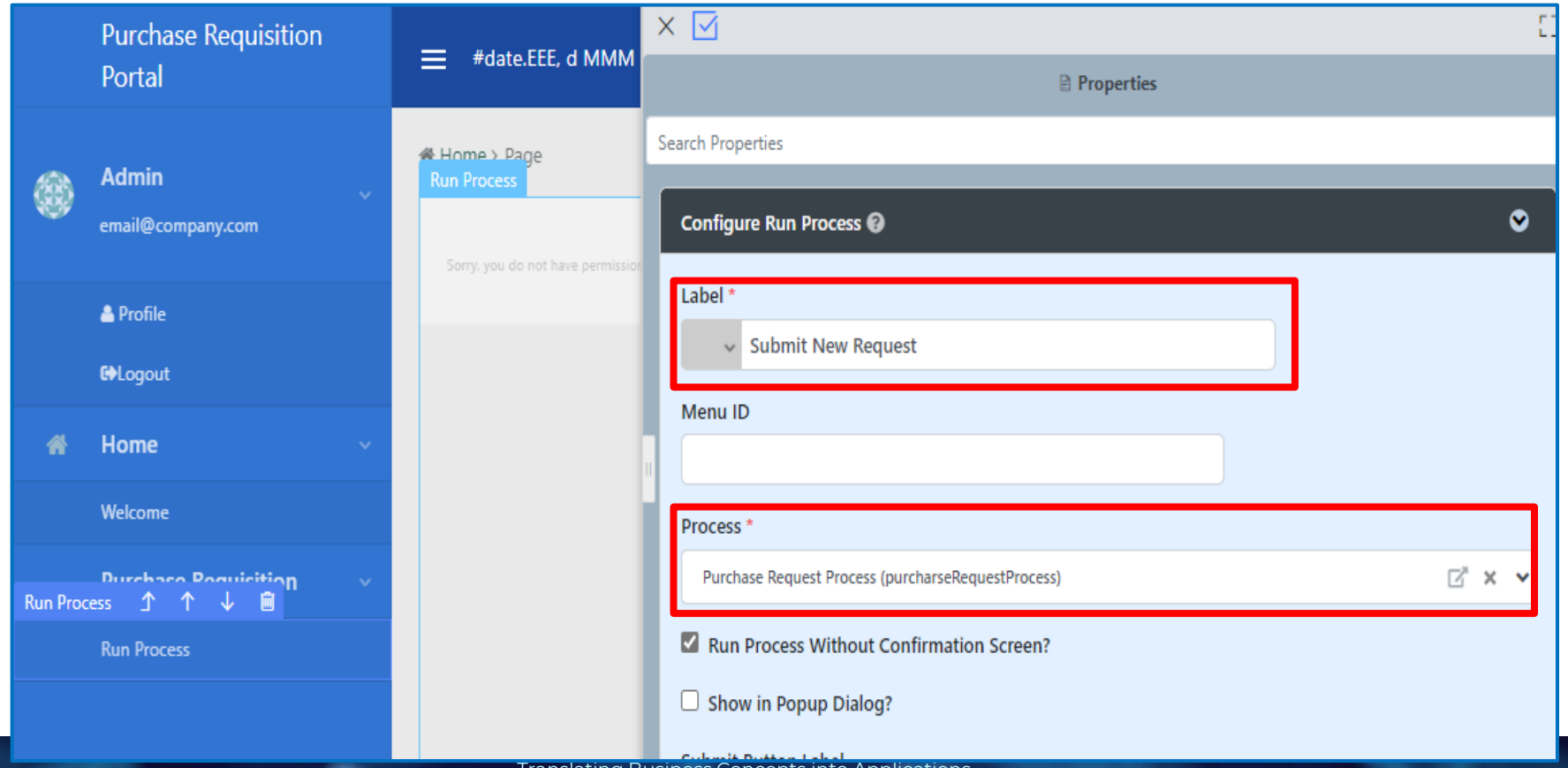
Add Run Process Into UI

- Drag-n-drop **Run Process** into **Purchase Requisition** category.



Integrate Run Process Into UI

- Set the appropriate label.
- Select Purchase Request Process in Process.



Add Inbox Into UI

- Add **Inbox** into **Purchase Requisition** category.
- Configure it to show **All Assignments**.

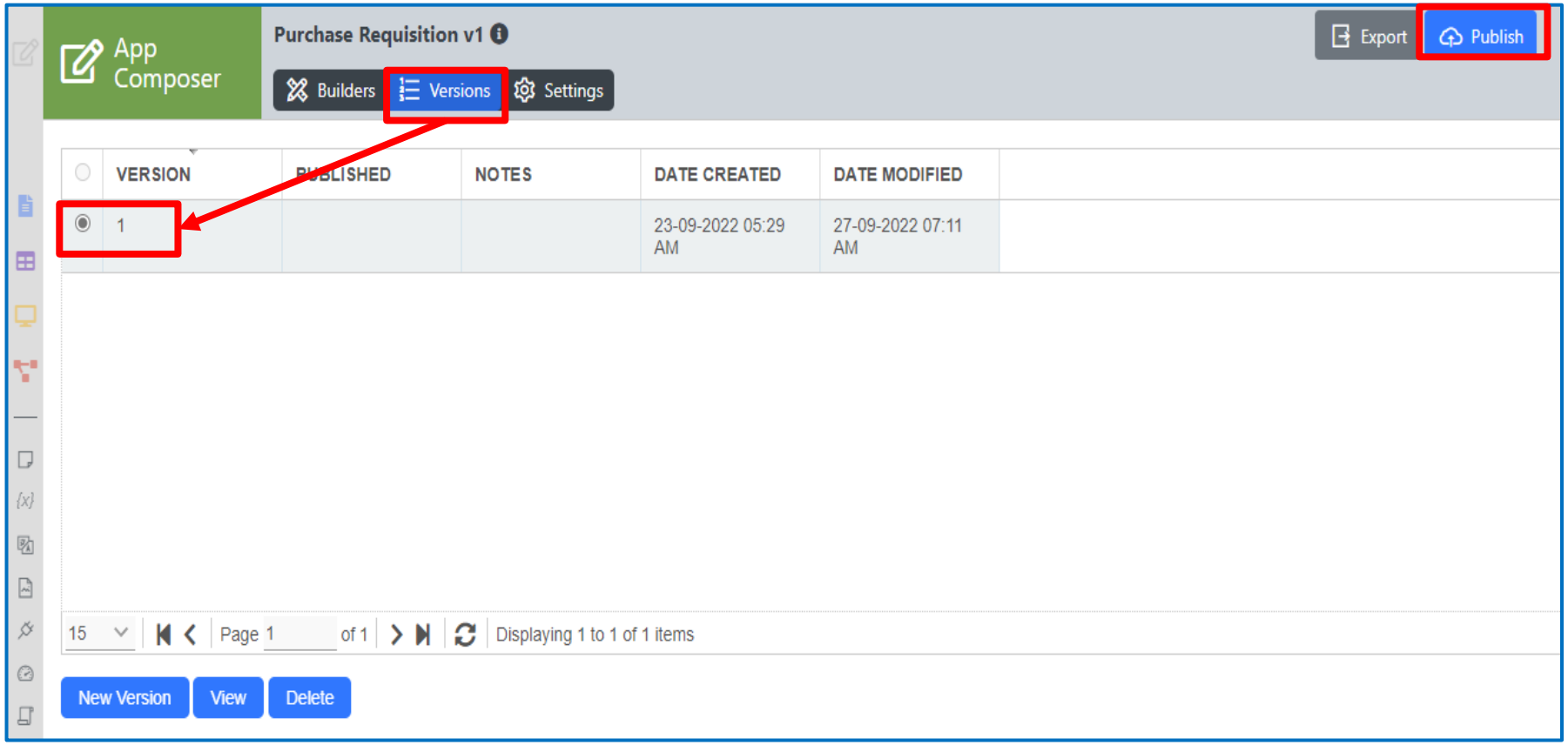
Publish the App

- With the UI created, it is now ready for prime time.
- End users can now access them through this interface when the App is made published.

Note: Only **ONE** App version can be made published at any point of time.

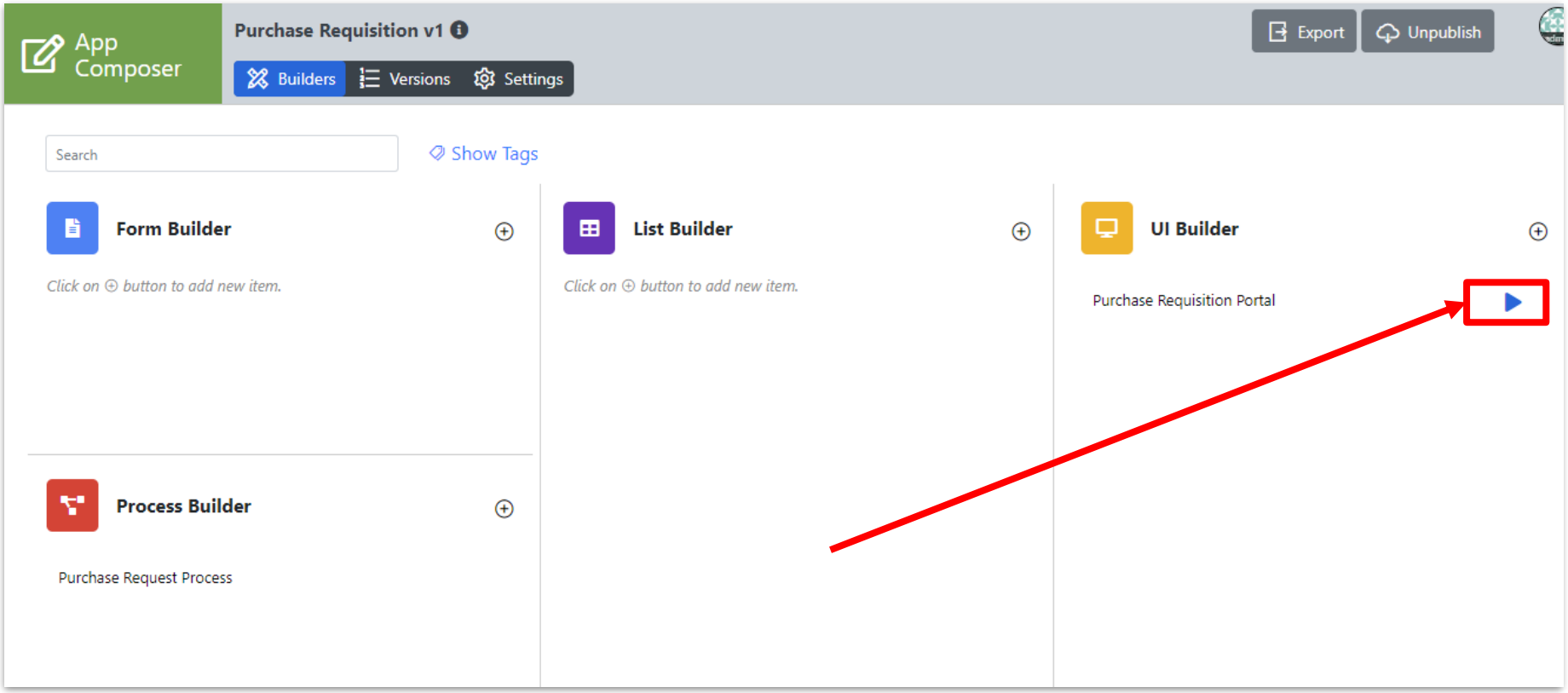
Publish the App

- Click on **Version** button, select the version to publish.



Launching the Published App

- Click on ▷under UI Builder pane.

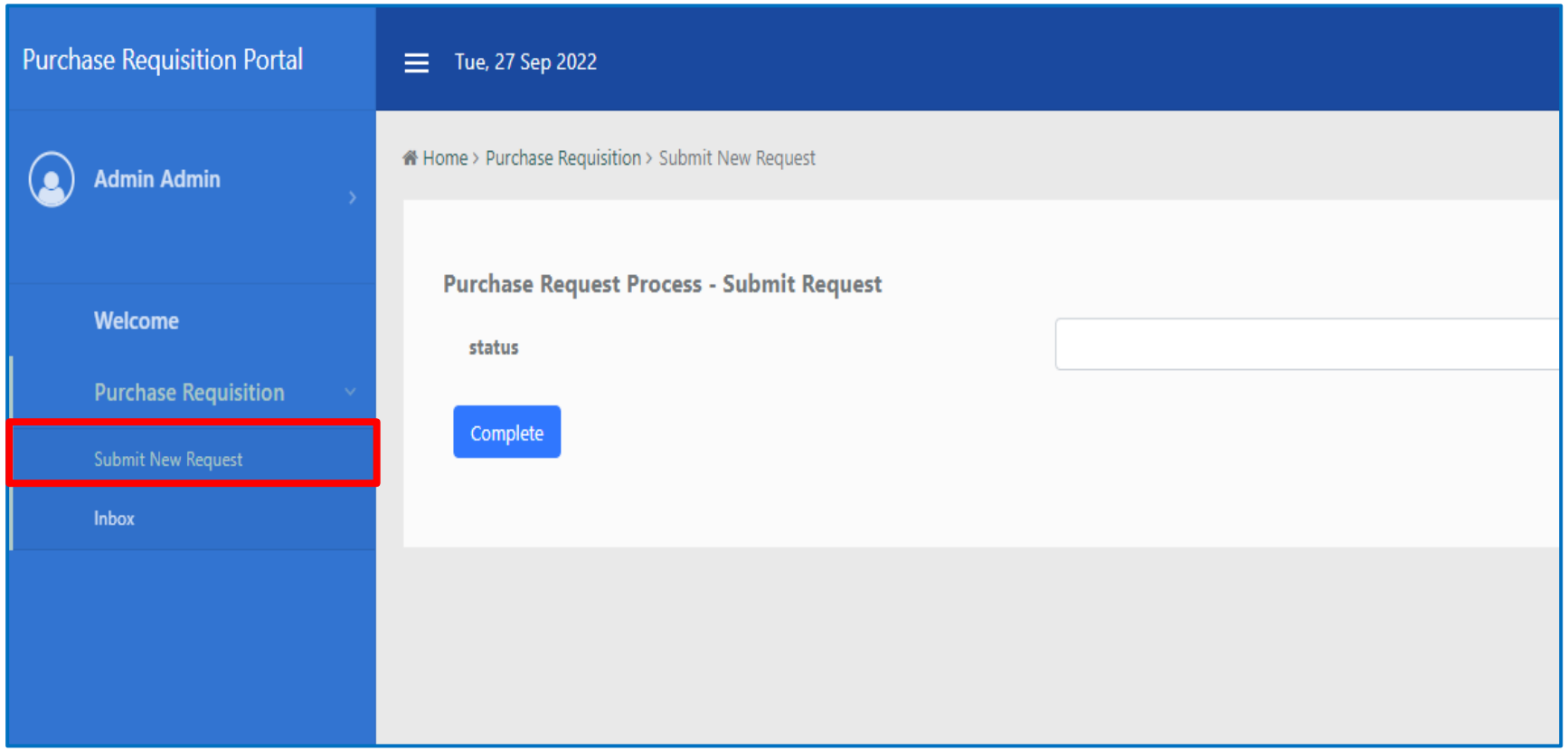


Launching the Published App

- One can also access it through the **App Center** itself.

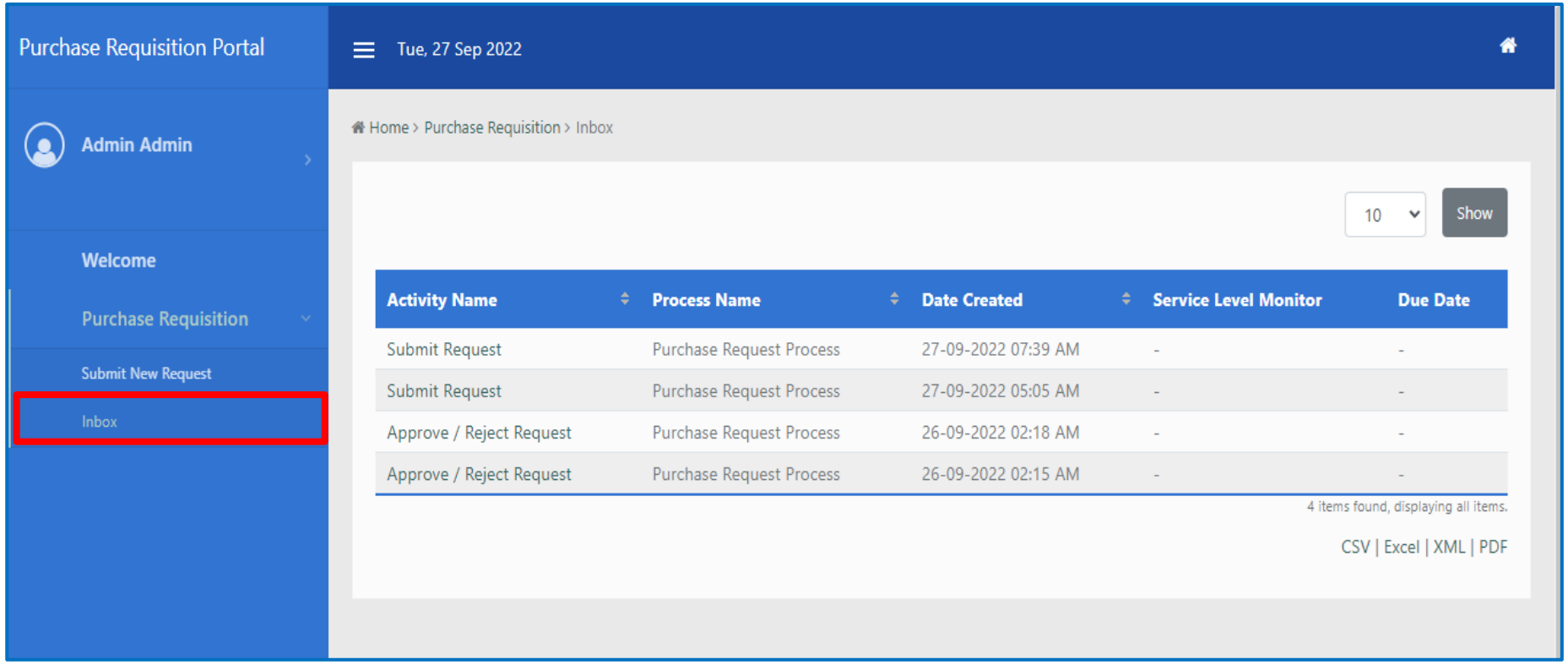
Run Through the Process

- With the UI created, run through the entire **Purchase Request Process** again by using the UI.



Run through the Process

- Access subsequent assignment through the Inbox



Good To Know

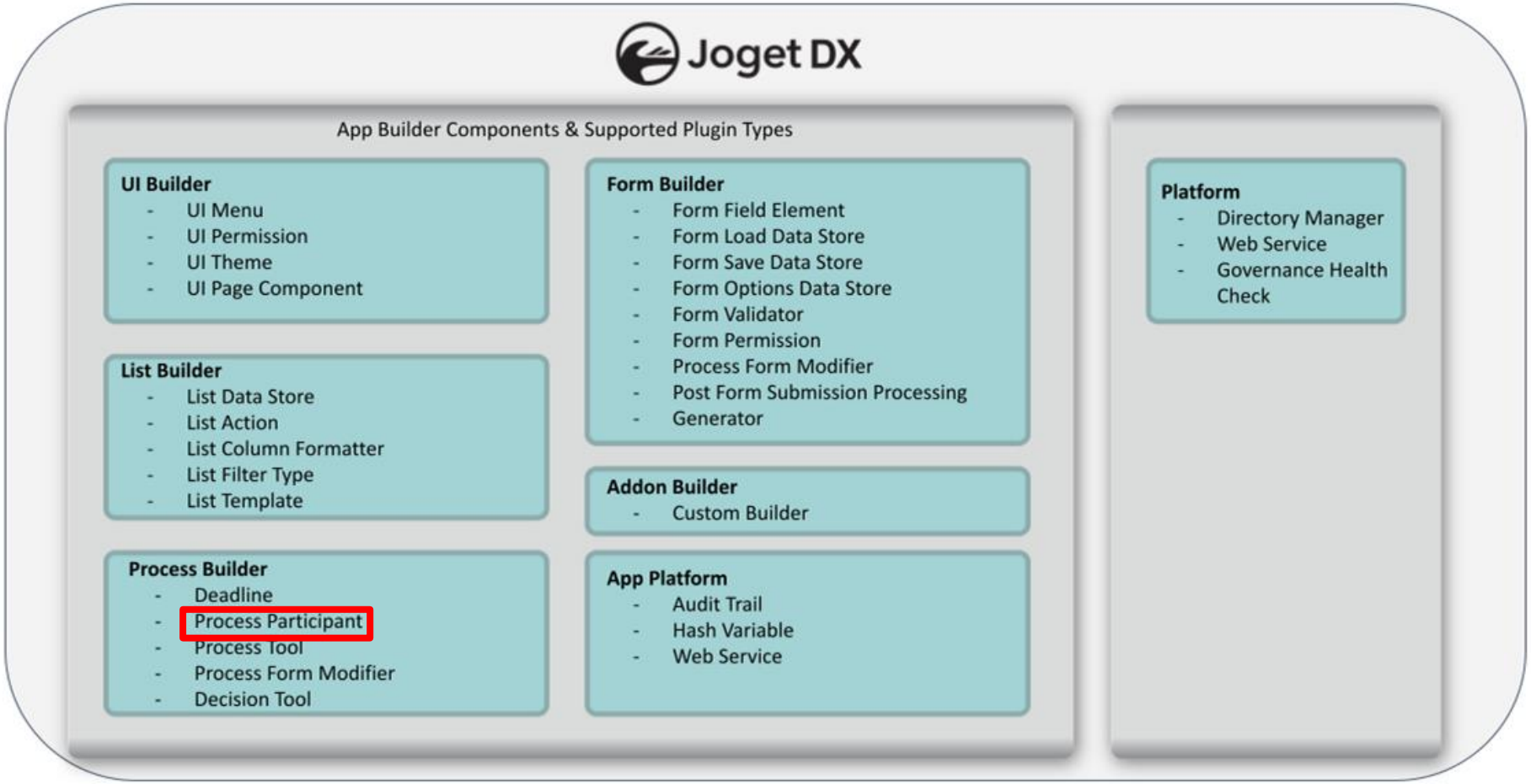
- At this point of time, the App is published, therefore end user would be able to access it.

Discussion

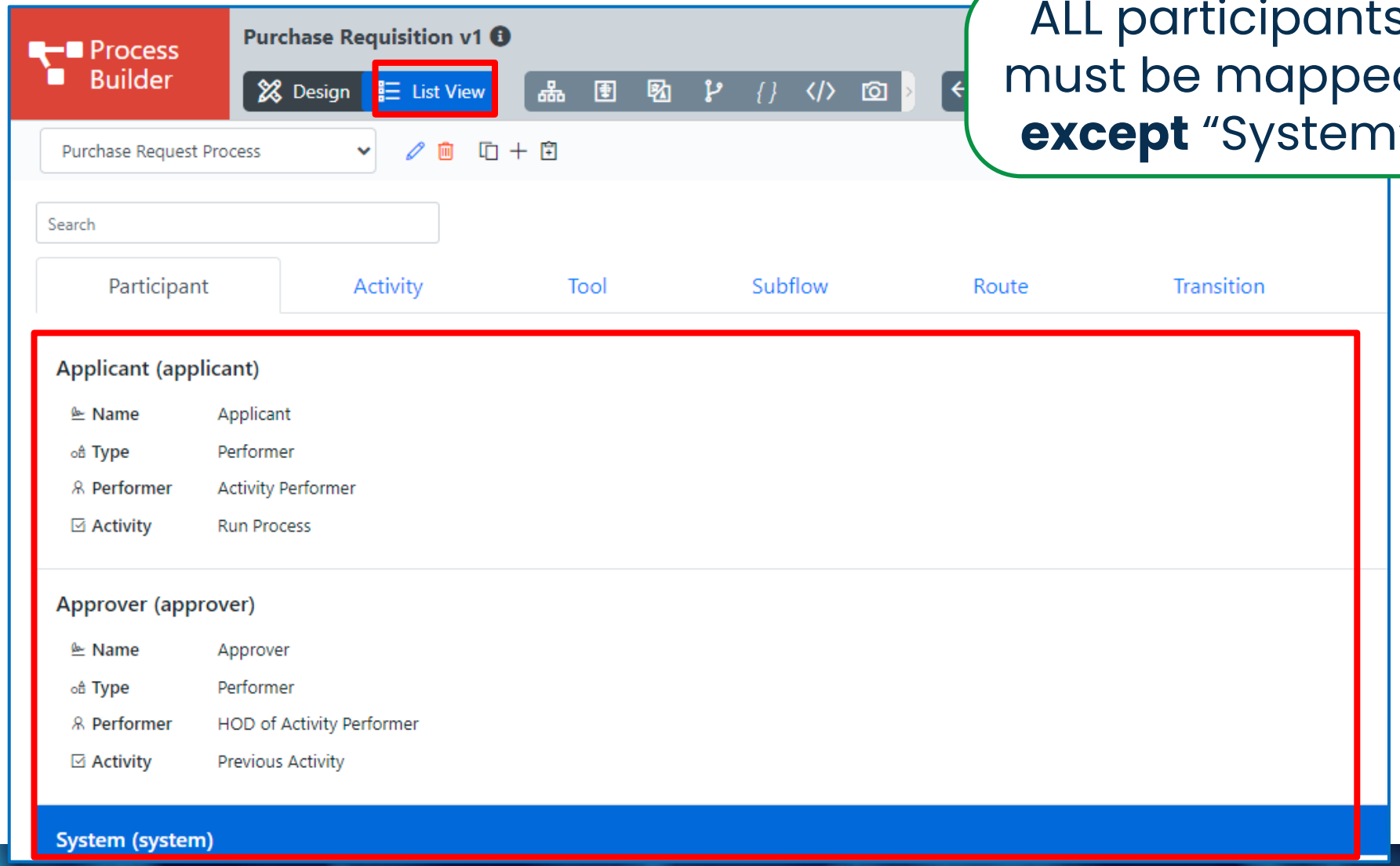
- Who is Applicant? Who is Approver?
  Have we actually defined who they really are?

Topic
Participant Mapping

Plugin Types



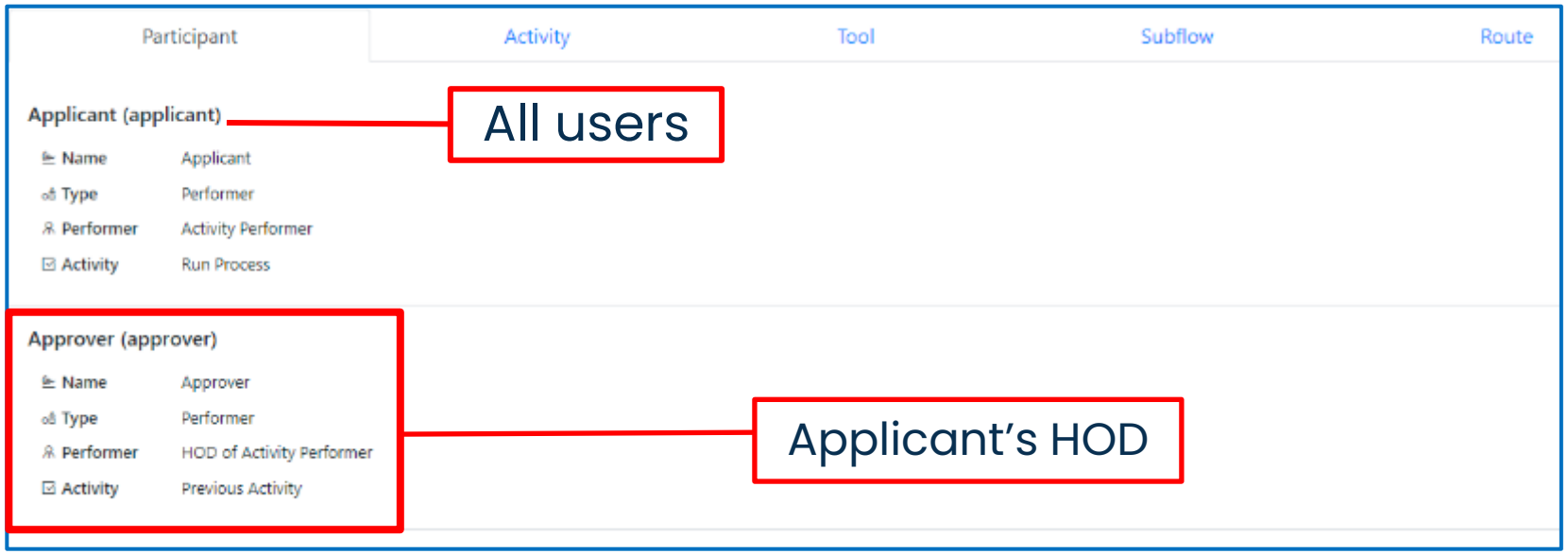
Where to map participants?



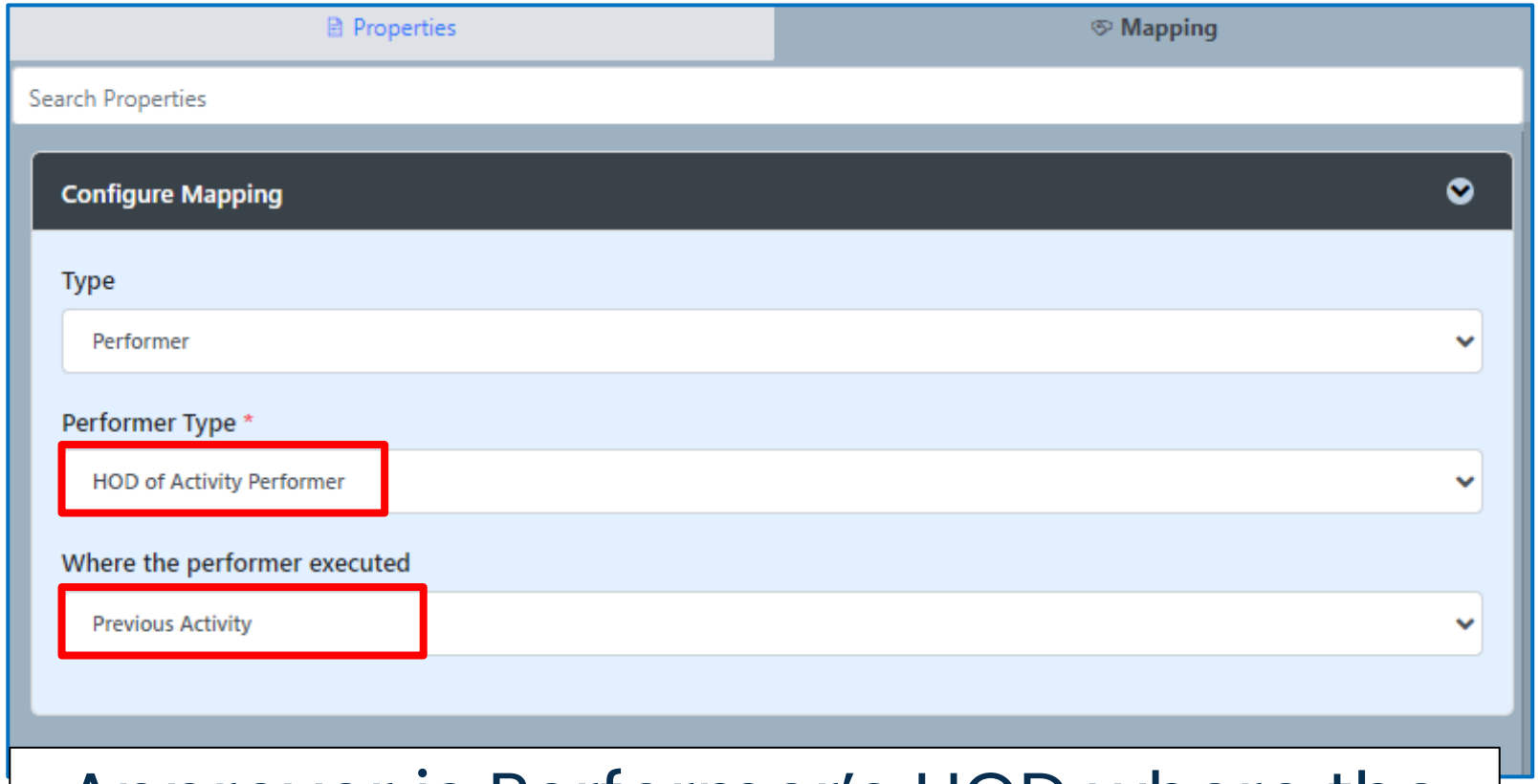
ALL participants must be mapped, **except** "System"

Participant Mapping

Let's assume that, applicant's HOD is responsible to approve any purchase requests made by the applicant.



Approver: Performer's HOD



Approver is Performer's HOD where the performer **executed** Previous Activity

Process Participant Mapping

Participant | Activity | Tool | Subflow | Route | Transition | Variables

Applicant (applicant)

Name	Applicant
Type	Performer
Performer	Activity Performer
Activity	Run Process

Approver (approver)

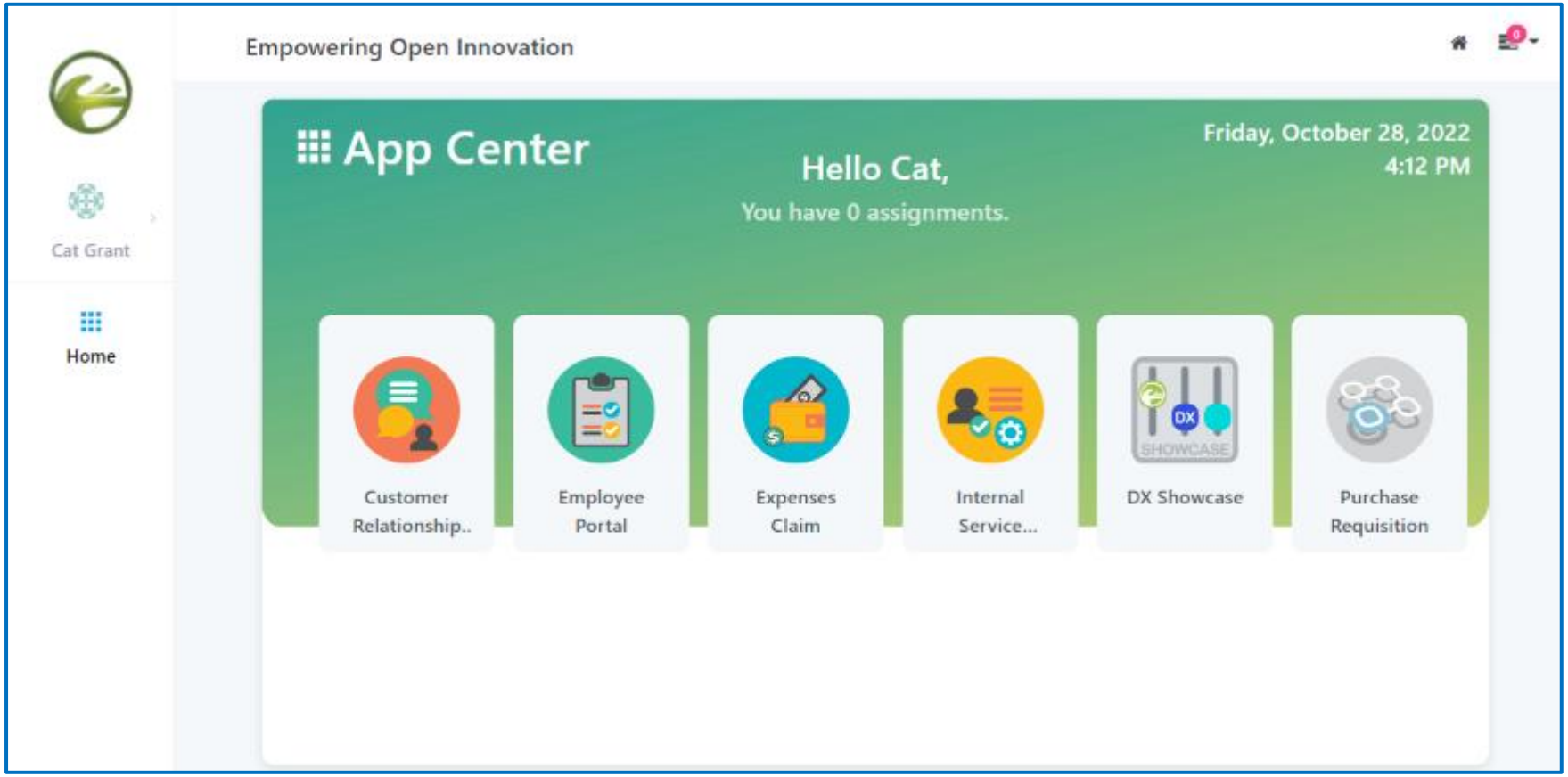
Name	Approver
Type	Performer
Performer	HOD of Activity Performer
Activity	Previous Activity

Test Run the Process

- With the participant mapping set, run the process again as the user “cat”, through the **Purchase Requisition Portal**.

Login As cat/password

- Open up App Center to access Purchase Requisition portal.



Important Notes

- All participants MUST be mapped.
- Do NOT choose activity from the same swimlane as the participant that is being mapped unless it is intended so.
- Choose only activity that has been completed.
- If NO user is found, it will be defaulted to the last performer. (Possibly, the current logged in user who had just completed the prior activity OR the process requester)
- If mapped to an inactive user, do delegate, re-assign, or change mapping and re-evaluate the assignment.

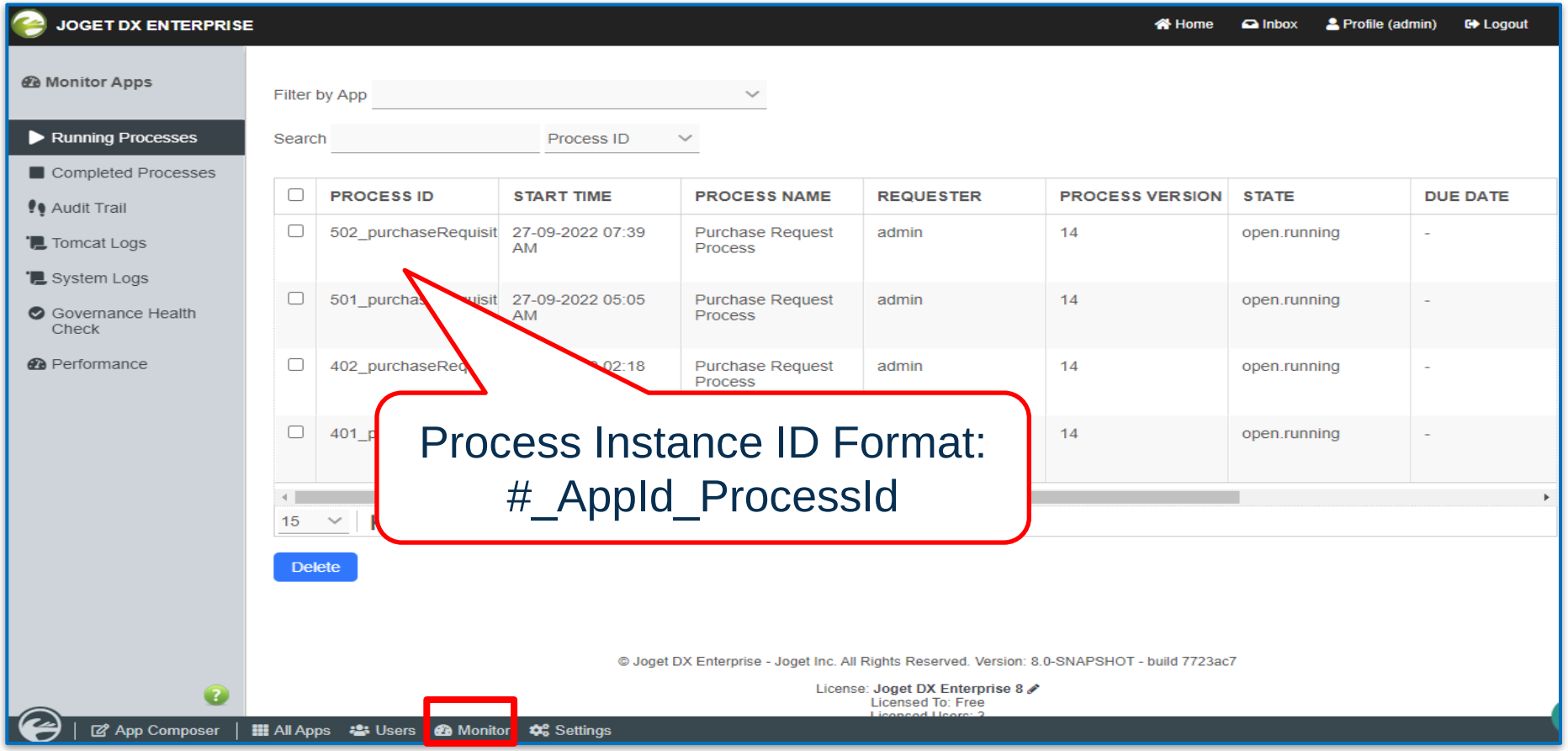
Topic
Process Monitoring

Monitoring Processes

- Allows you to monitor the status of processes, both **running** and **archived**.
- Allows you to view an **audit trail** of process actions.
- Captures process data to allow for the generation of reports.

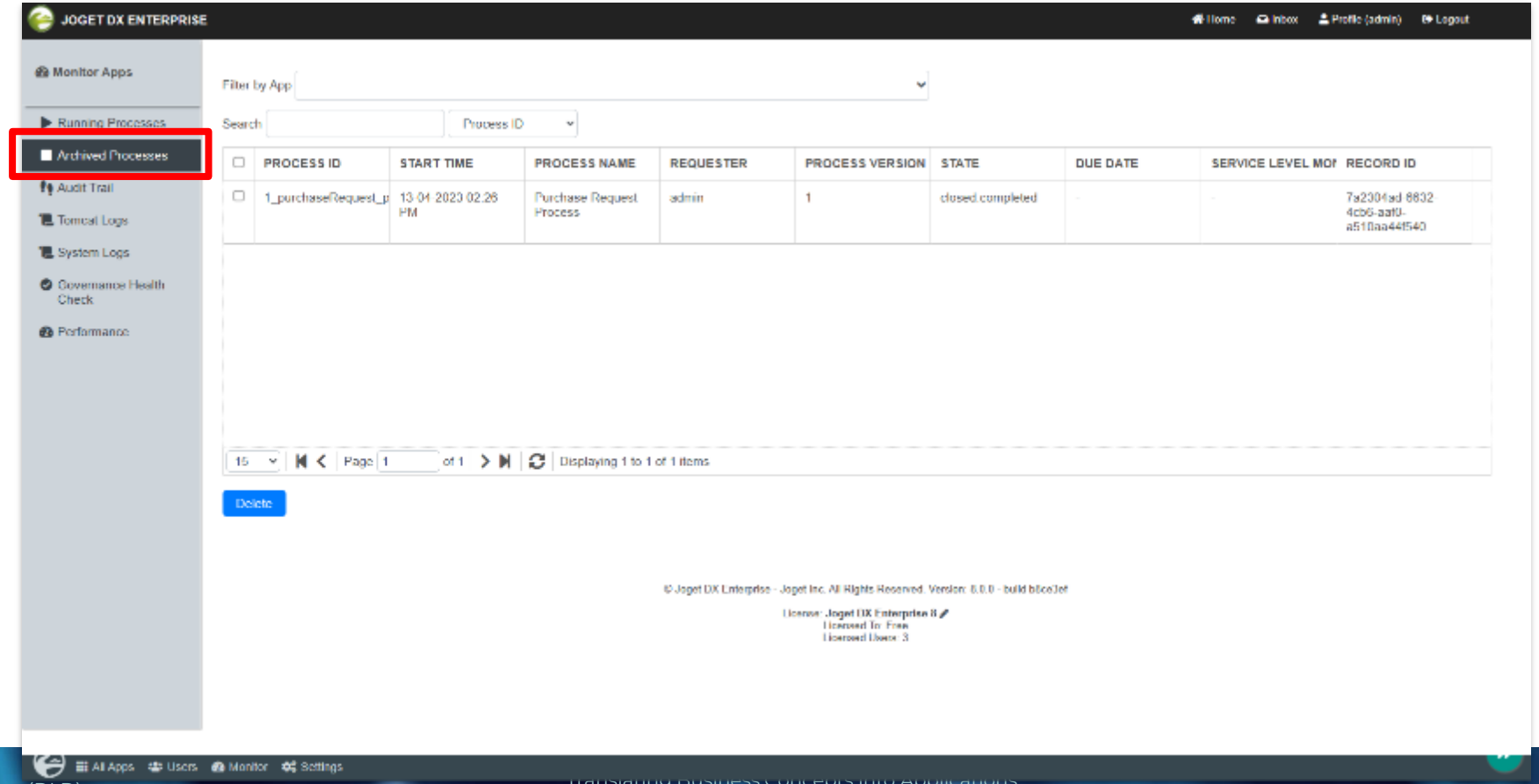
Monitor Running Processes

- Click on Monitor from the Admin Bar to access.



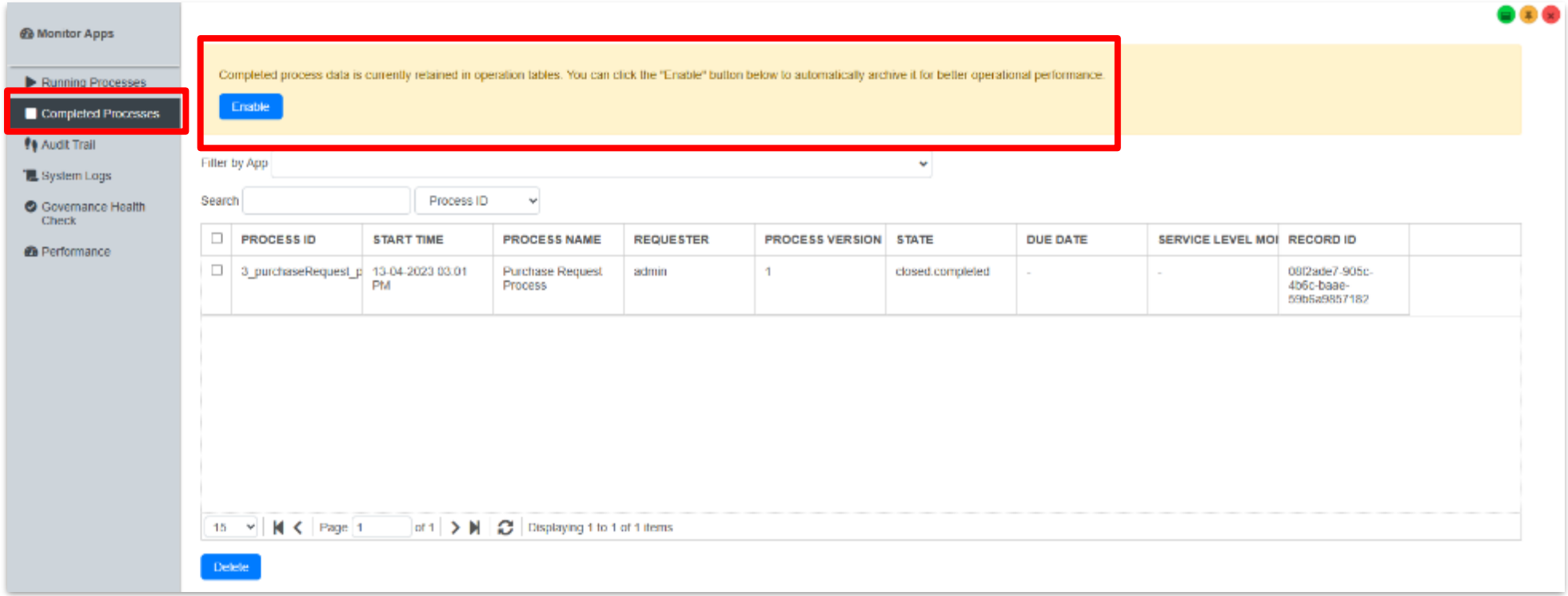
Monitor Archived Processes

- Click on “Archived Processes” to see completed process instances.



Retain Process on Completion

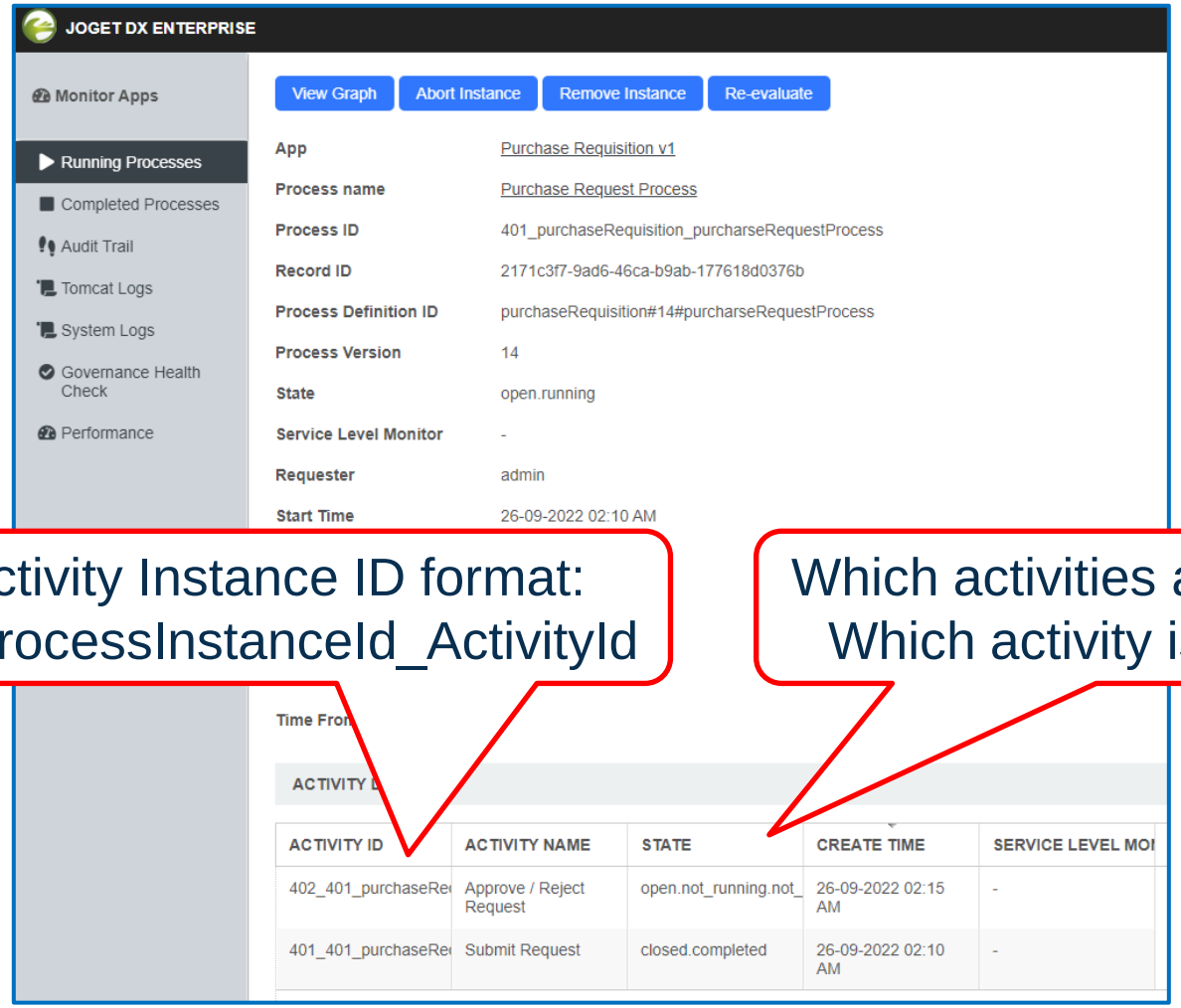
- Click on “Completed Processes” to see completed process instances.



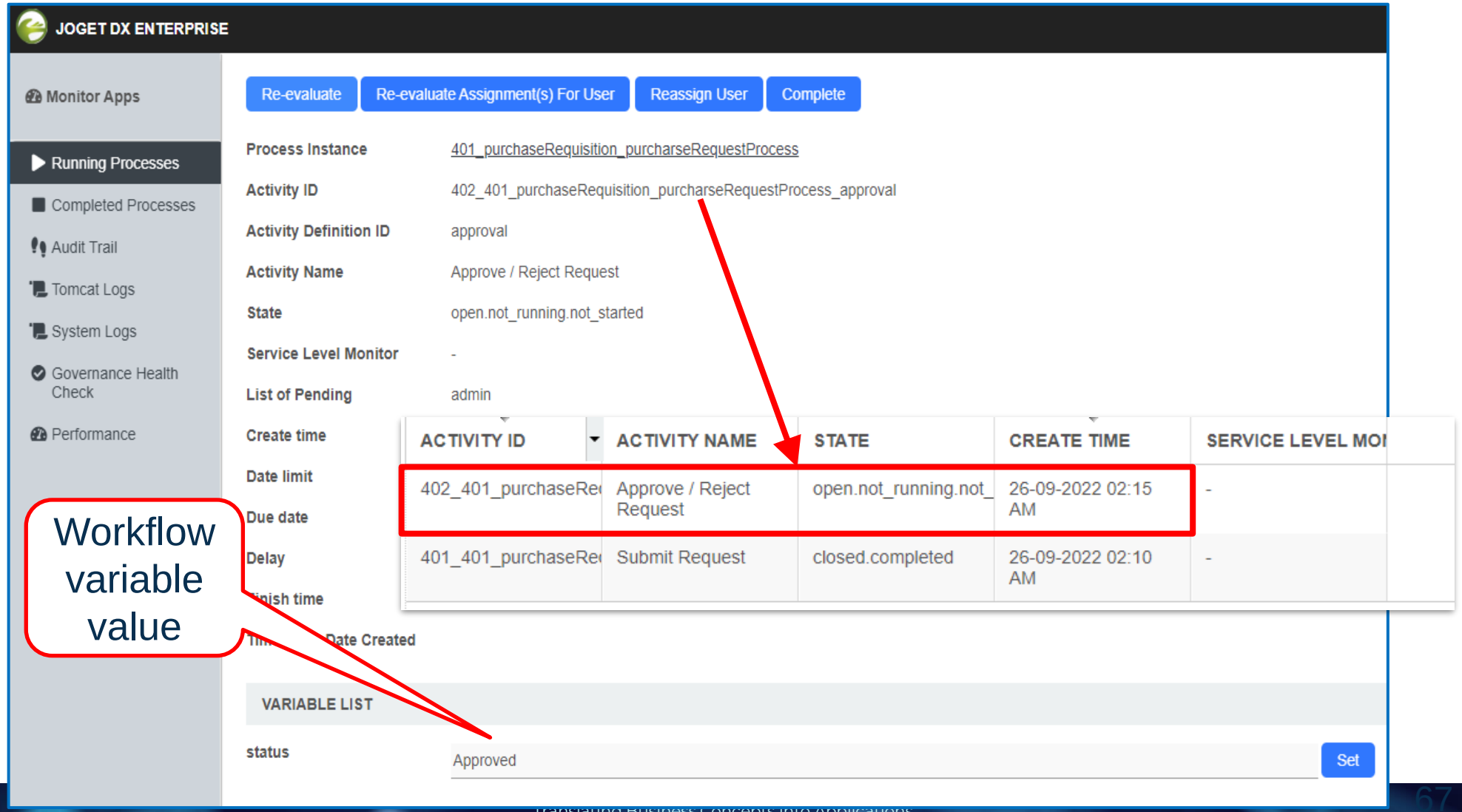
View the Details of a Running Process Instance

- From the list of running processes, **click into any instance**.
- **Details on the current state** of the selected process instance will be shown.

View the Details of a Running Process Instance



Activity Instance ID format:
#_ProcessInstanceId_ActivityId

Which activities are completed?
Which activity is active now?

View the Details of an Activity